Advanced Representation Learning
(고급표현학습)

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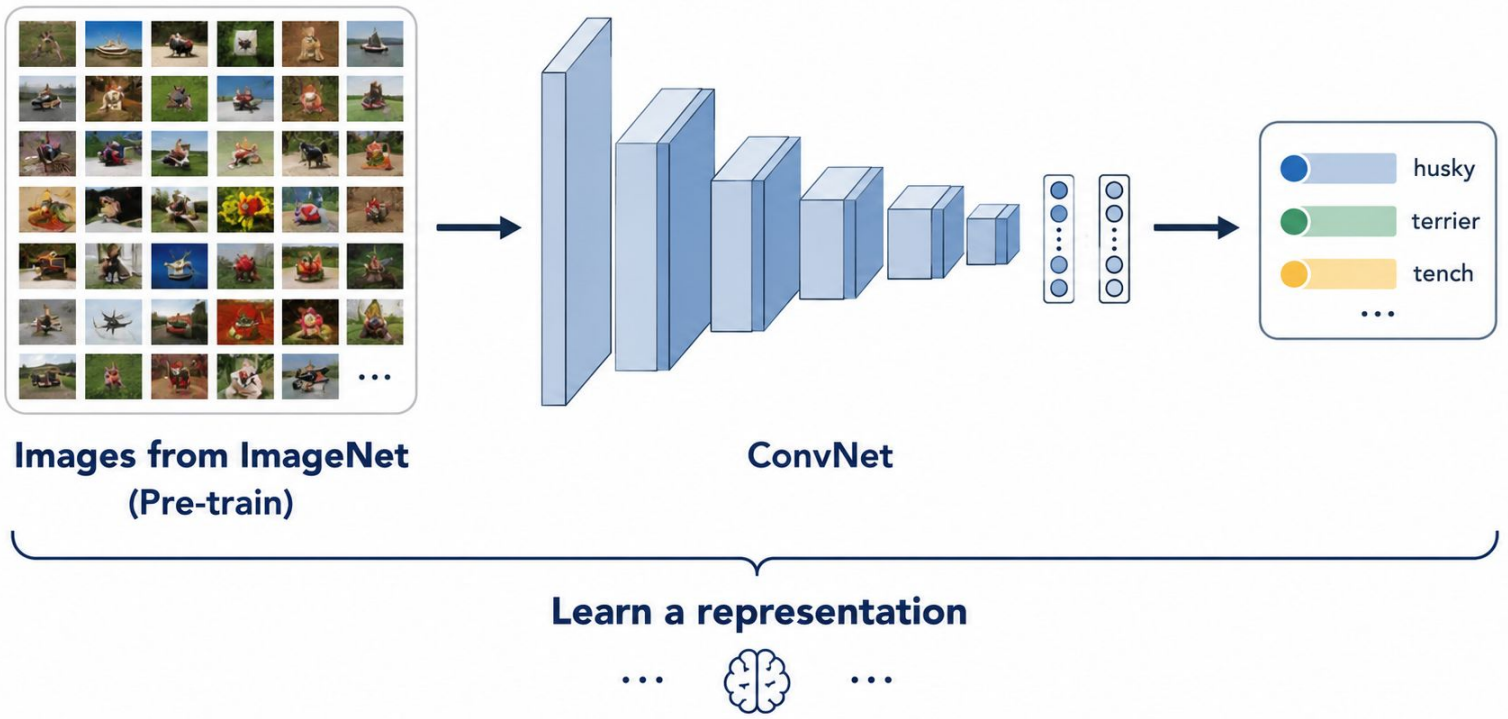
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Why Self-Supervised Learning?

Success Story of Supervision: Pre-training

- Features from networks pre-trained on ImageNet can be used for a variety of different downstream tasks.



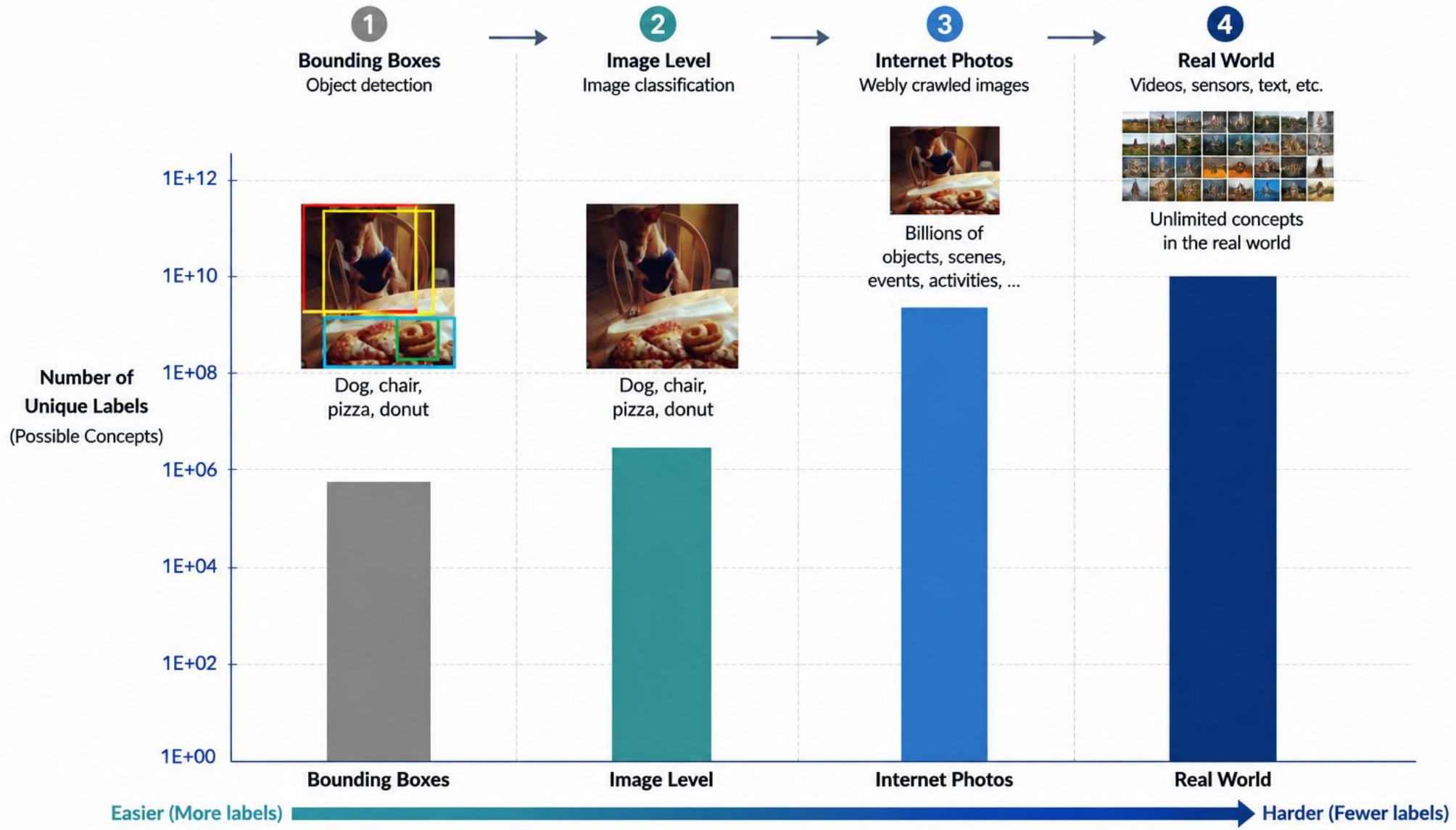
Success Story of Supervision: Recipe for Good Solutions

- Pre-train on a large supervised dataset.
- Collect a dataset of "supervised" images
- Train a ConvNet

Limitations of Supervised Learning: Challenges in Labeling

- Getting "real" labels is difficult and expensive
  - ImageNet with 14M images took 22 human years.
- Obtain labels using a "semi-automatic" process
  - Hashtags
  - GPS locations

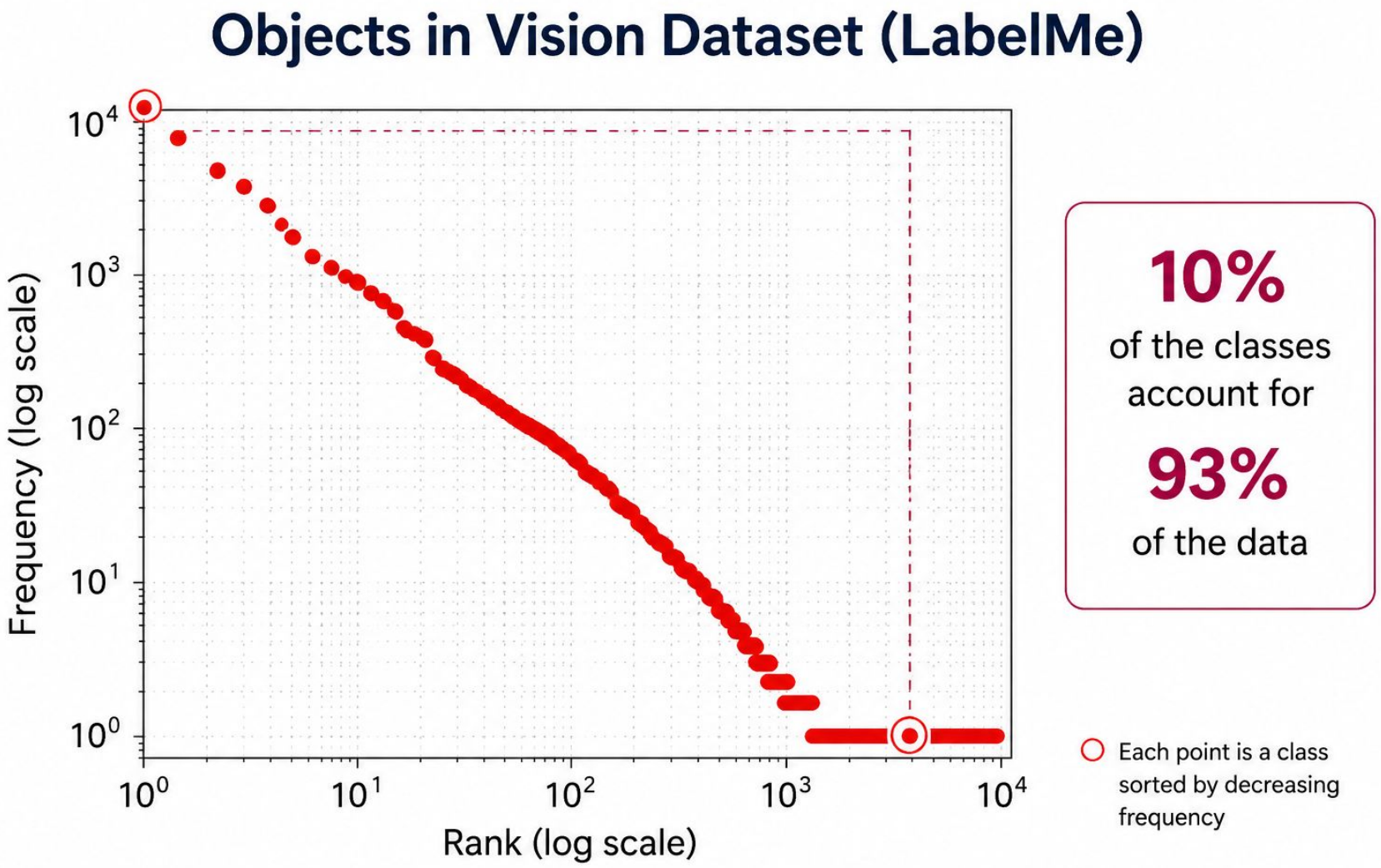
Can We Get Labels for All Data?



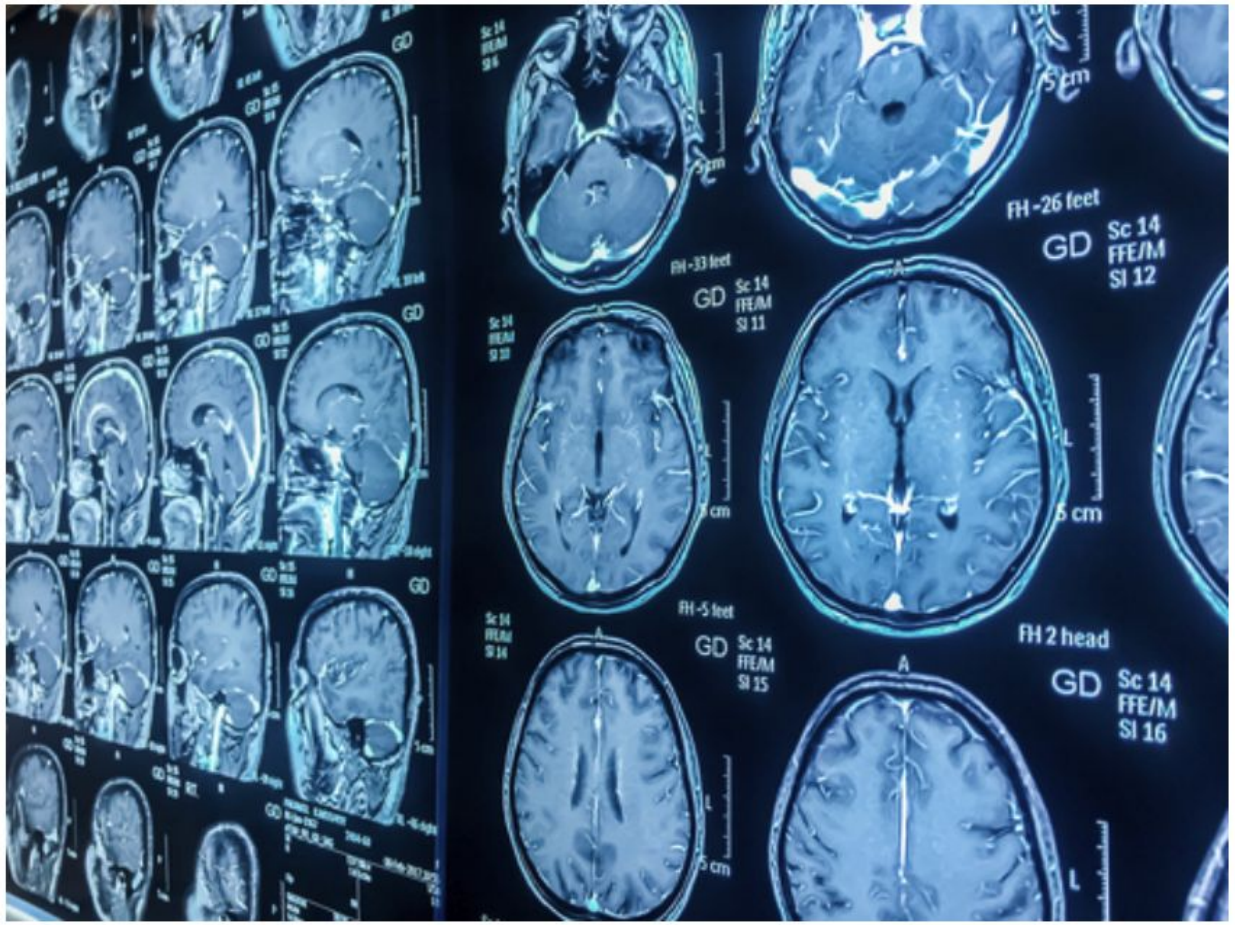
Can We Get Labels for All Data?

- What about complex concepts?
  - Video?
- Labelling cannot scale to the size of the data we generate

Rare Concepts?



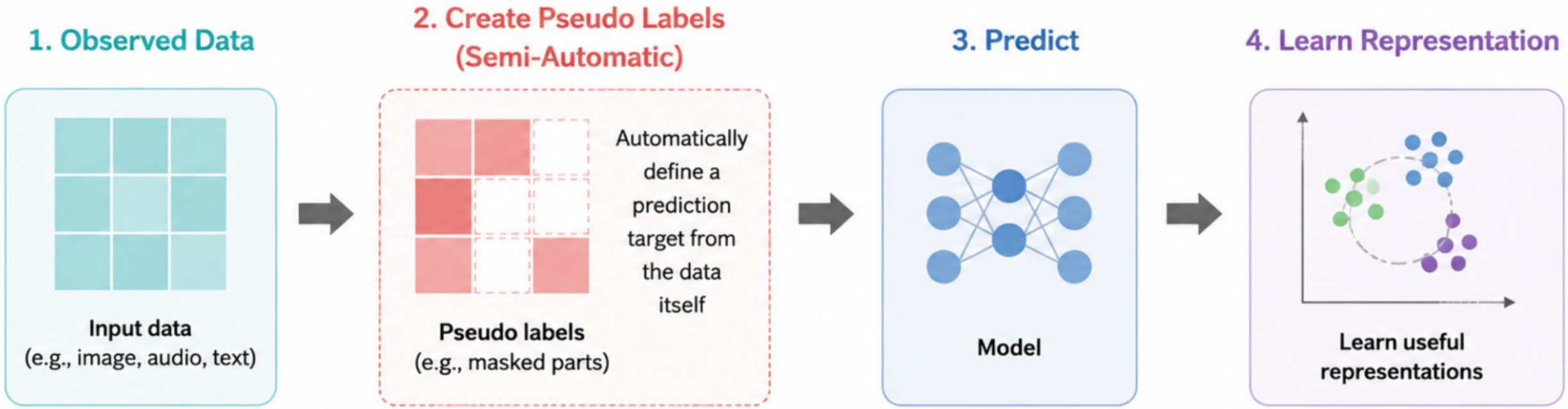
Different Domains?



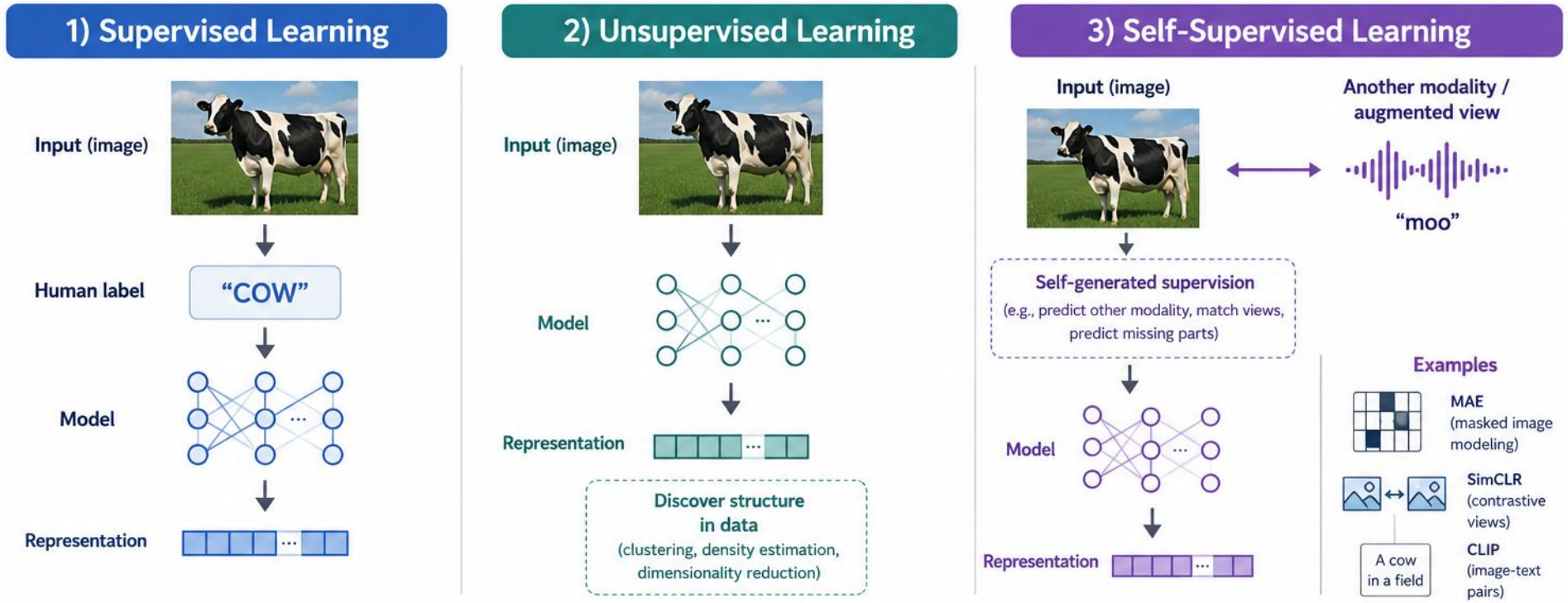
What is Self-Supervised Learning?

What is "Self" Supervision?

- Obtain "labels" from the data itself by using a "semi-automatic" process.
  - Predict part of the data from other parts.



What is "Self" Supervision?

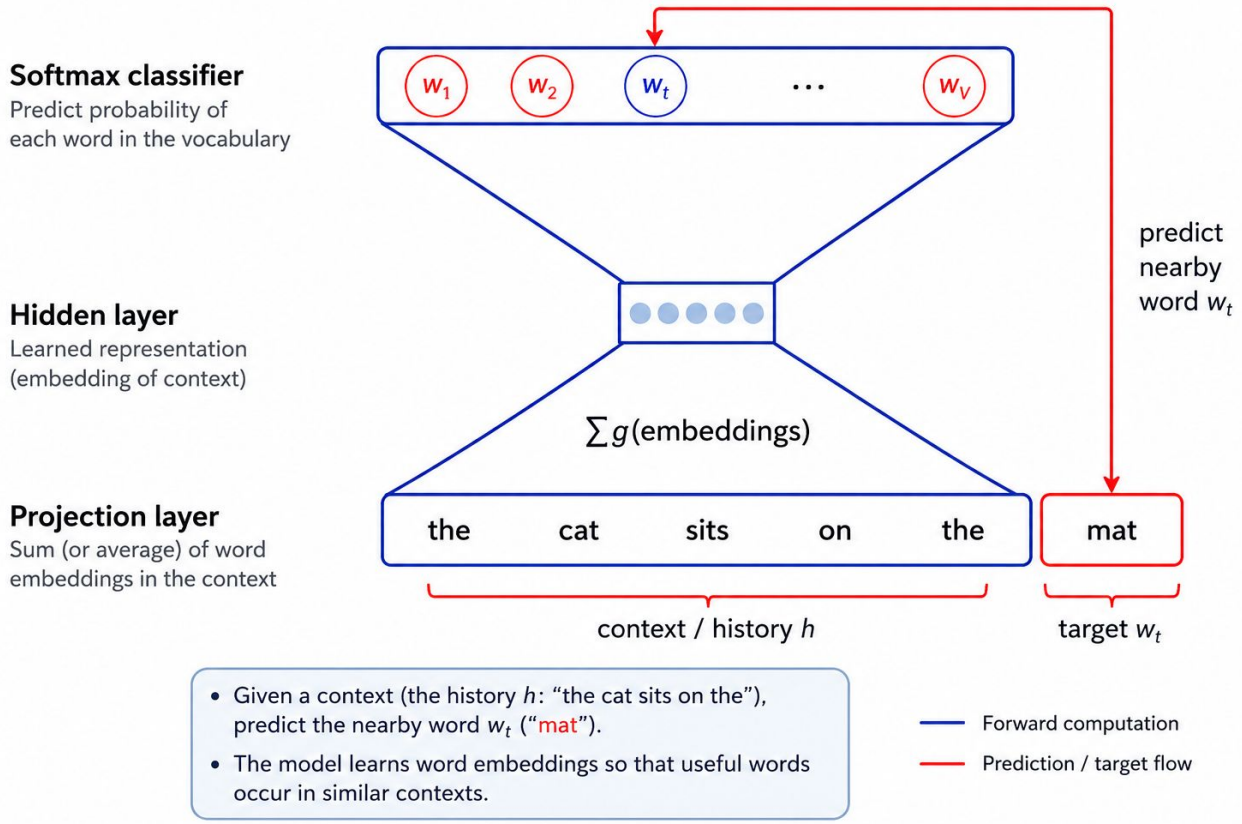


	Supervised	Unsupervised	Self-Supervised
Labels	Human provided (expensive)	No labels	Automatically generated from data
Source of supervision	Human annotations	None	Data itself (pretext tasks)
Cost	High	Low	Low
Scalability	Limited	High	Very high
Learned representations	Task-specific	Limited semantic meaning	Rich, transferable representations

Self-supervised learning creates supervision signals **directly from the data itself**, enabling scalable learning of meaningful representations.

Success of Self-Supervised Learning in NLP

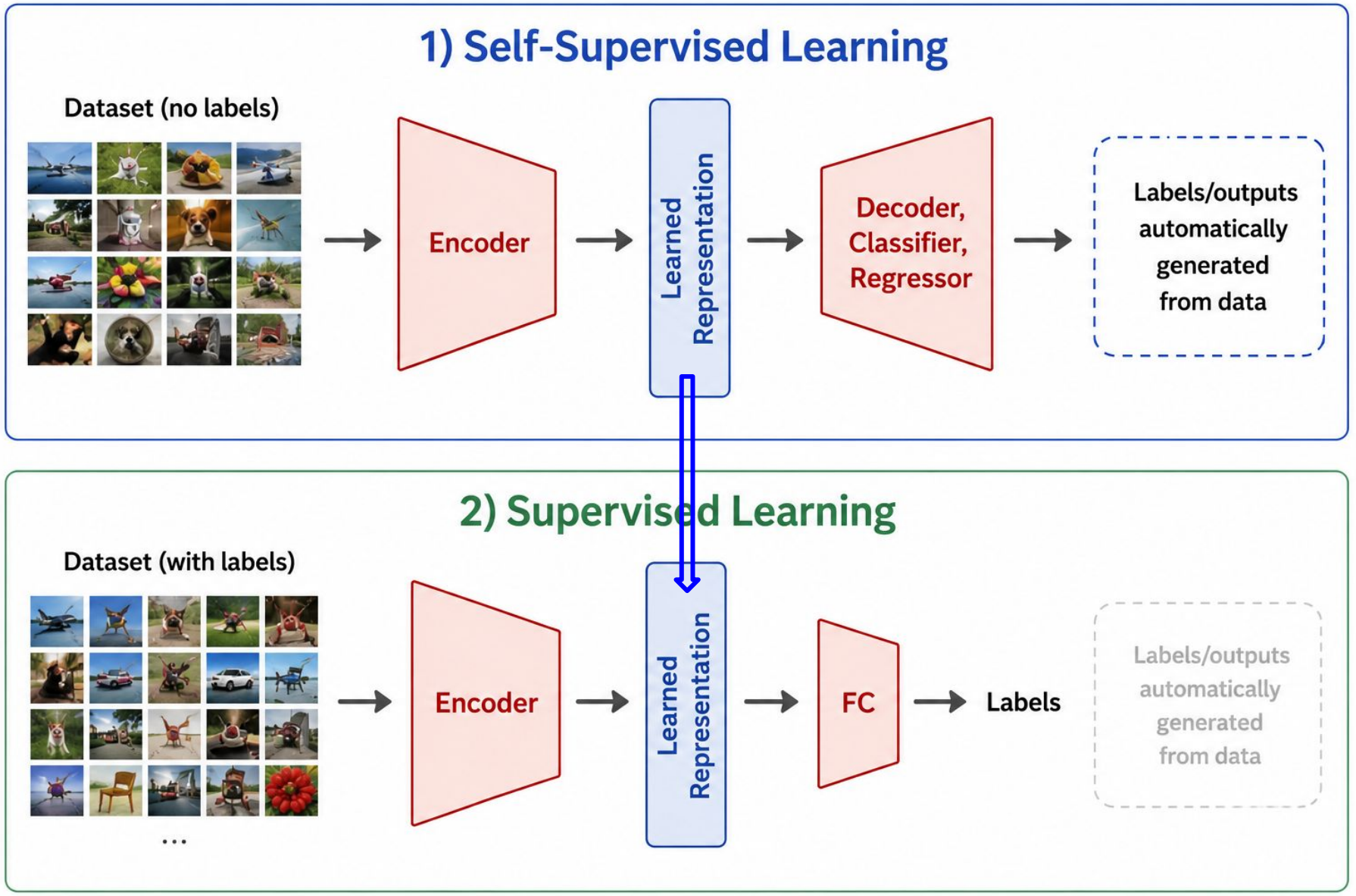
- Ex. Word2vec
- Fill in the blanks is a powerful signal to learn representations
- Sentence/Word representations: BERT - Devlin et al., 2018



Why Self Supervision?

- Helps us learn using observations and interactions
- Does not require exhaustive annotation of concepts
- Leverage multiple modalities or structure in the domain

Self-Supervised Learning Paradigm



Pretext Task
- Define a task based on the data itself
- No manual annotation
- Could be considered an unsupervised task;
- but we learn with supervised learning objectives, e.g., classification or regression.

Downstream Task
- The application you care about
- You do not have large datasets
- The dataset is labeled

Self-Supervised Pretext Tasks

- Example: learn to predict image transformations / complete corrupted images

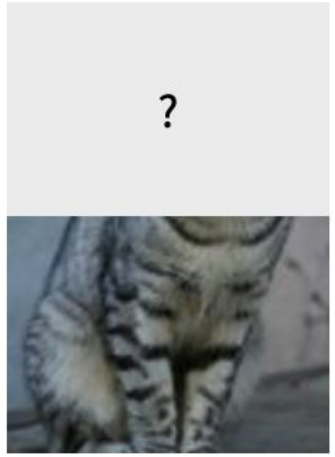
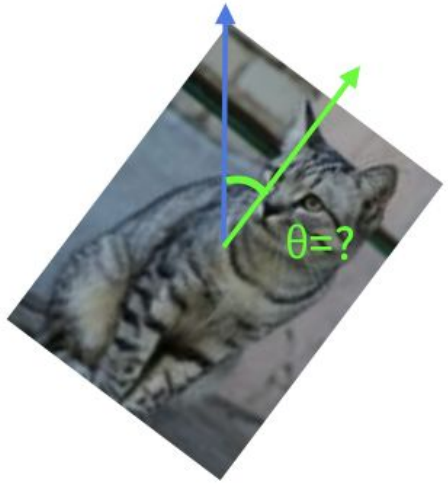


image completion



rotation prediction



"jigsaw puzzle"

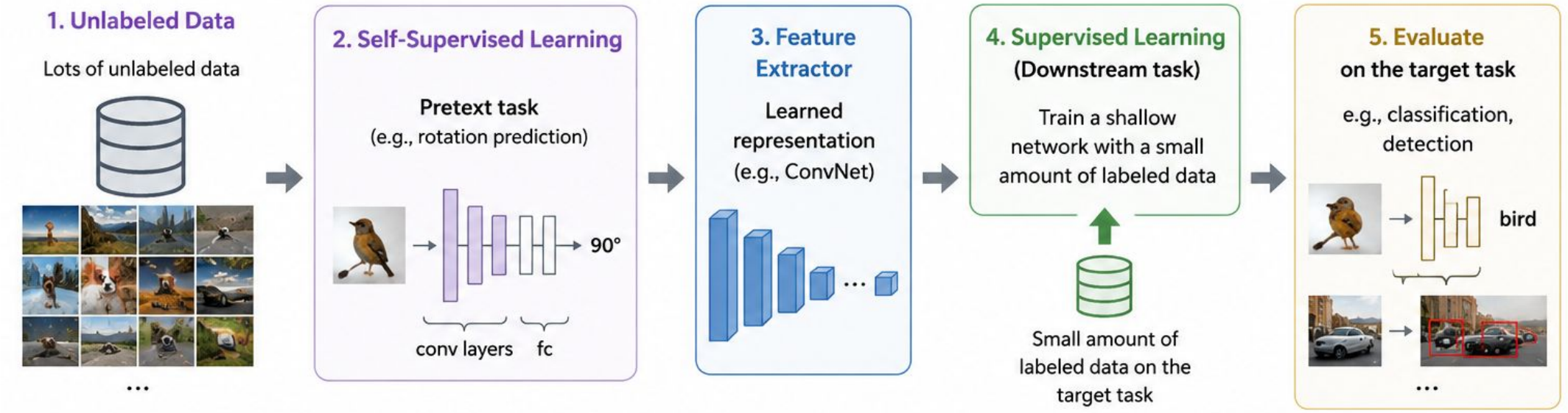


colorization

1. Solving the pretext tasks allow the model to learn good features.
2. We can automatically generate labels for the pretext tasks.

How to Evaluate a Self-Supervised Learning Method?

- Learn good feature extractors from self-supervised pretext tasks, e.g., predicting image rotations.
- Attach a shallow network on the feature extractor; train the shallow network on the target task with small amount of labeled data.

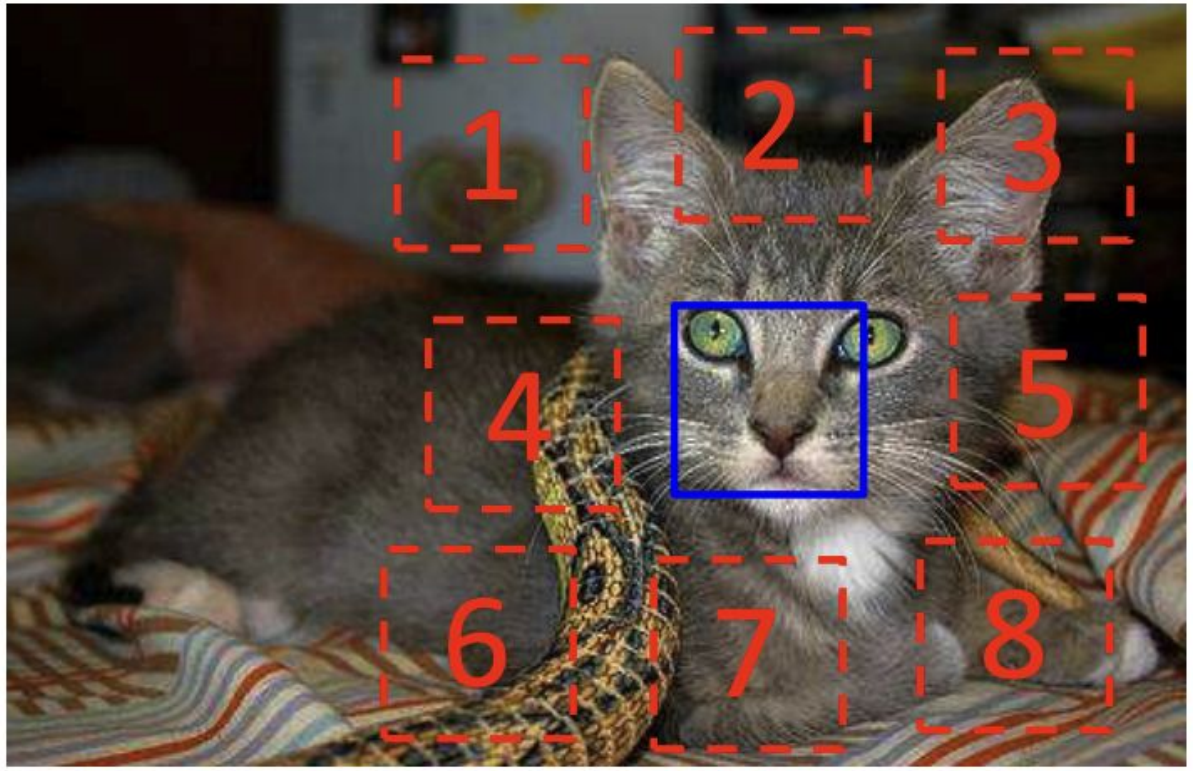


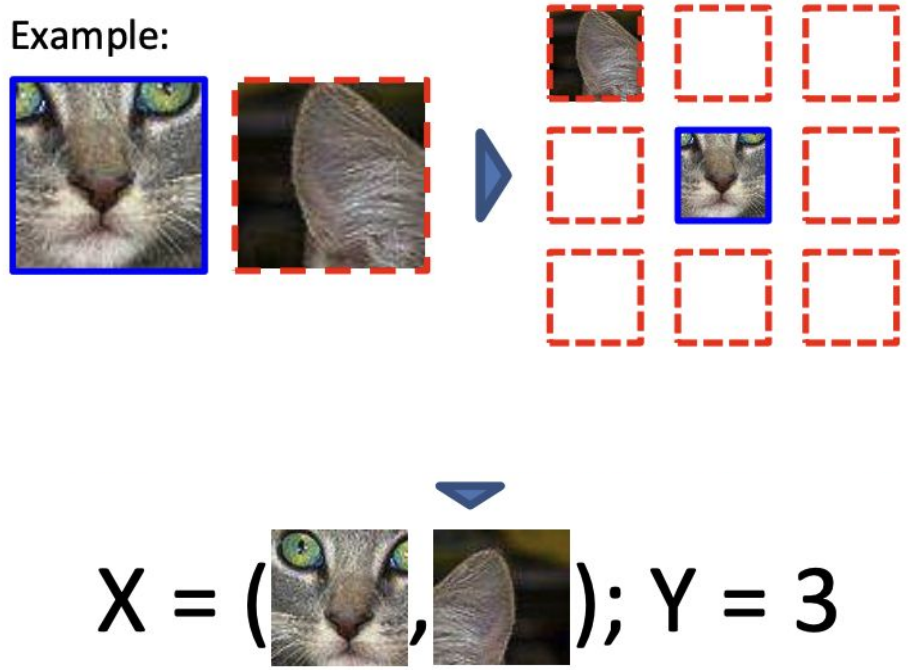
Pretext Tasks

Pretext Task

- Self-supervised task used for learning representations
- Often, not the "real" task (like image classification) we care about

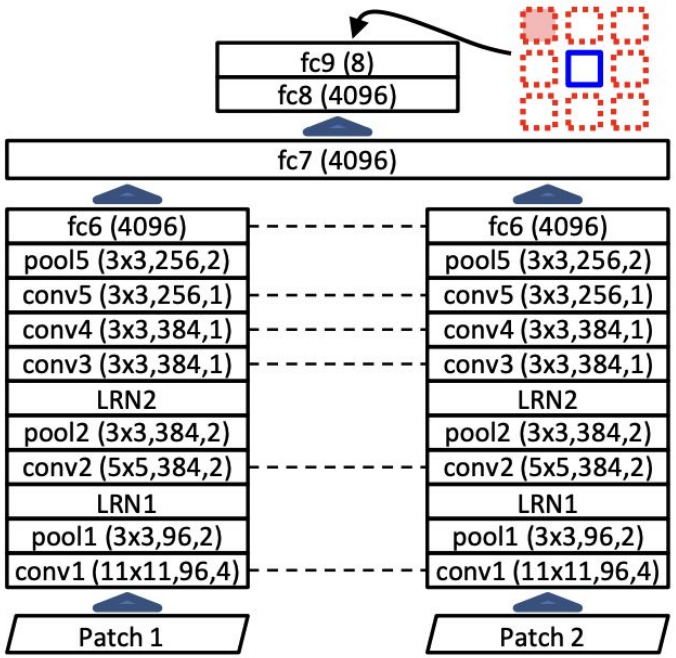
Pretext Task1: Relative Position of Patches



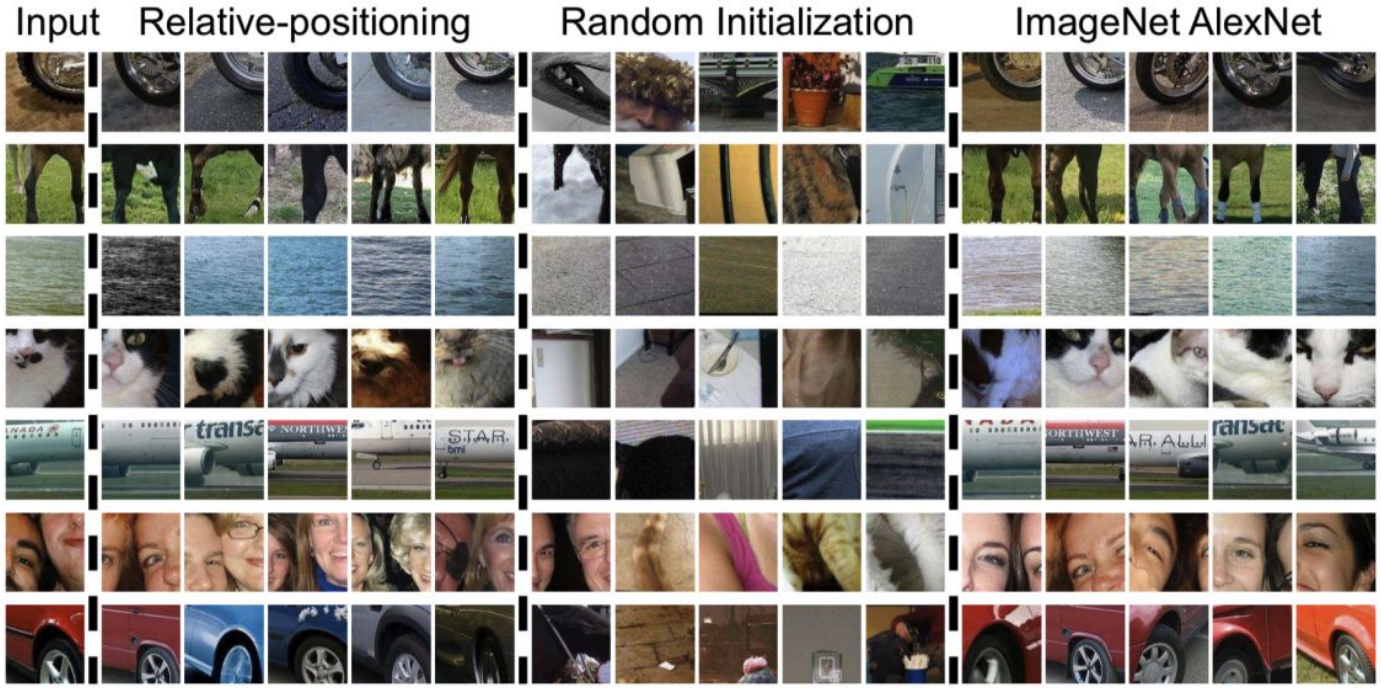


Doersch et al., ICCV 2015, Unsupervised visual representation learning by context prediction

Pretext Task1: Relative Position of Patches







Nearest Neighbors in features

Doersch et al., ICCV 2015, Unsupervised visual representation learning by context prediction

Pretext Task1: Relative Position of Patches

- Self-supervised pretraining significantly improves downstream object detection performance, even without human annotations.

VOC-2007 Test	aero	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	mbike	person	plant	sheep	sofa	train	tv	mAP
DPM-v5[17]	33.2	60.3	10.2	16.1	**27.3**	54.3	58.2	23.0	20.0	24.1	26.7	12.7	58.1	48.2	43.2	12.0	21.1	36.1	46.0	43.5	33.7
[8] w/o context	52.6	52.6	19.2	25.4	18.7	47.3	56.9	42.1	16.6	41.4	41.9	27.7	47.9	51.5	29.9	20.0	41.1	36.4	48.6	53.2	38.5
Regionlets[55]	54.2	52.0	20.3	24.0	20.1	55.5	68.7	42.6	19.2	44.2	**49.1**	26.6	57.0	54.5	43.4	16.4	36.6	37.7	**59.4**	52.3	41.7
Scratch-R-CNN[2]	49.9	60.6	24.7	23.7	20.3	52.5	64.8	32.9	20.4	43.5	34.2	29.9	49.0	60.4	47.5	28.0	42.3	28.6	51.2	50.0	40.7
Scratch-Ours	52.6	60.5	23.8	24.3	18.1	50.6	65.9	29.2	19.5	43.5	35.2	27.6	46.5	59.4	46.5	25.6	42.4	23.5	50.0	50.6	39.8
Ours-projection	58.4	62.8	**33.5**	27.7	24.4	**58.5**	68.5	41.2	**26.3**	**49.5**	42.6	**37.3**	**55.7**	**62.5**	49.4	**29.0**	**47.5**	28.4	54.7	56.8	45.7
Ours-color-dropping	**60.5**	**66.5**	29.6	**28.5**	26.3	56.1	**70.4**	**44.8**	24.6	45.5	45.4	35.1	52.2	60.2	**50.0**	28.1	46.7	**42.6**	54.8	**58.6**	**46.3**
Ours-Yahoo100m	56.2	63.9	29.8	27.8	23.9	57.4	69.8	35.6	23.7	47.4	43.0	29.5	52.9	62.0	48.7	28.4	45.1	33.6	49.0	55.5	44.2
Ours-VGG	63.6	64.4	42.0	42.9	18.9	67.9	69.5	65.9	28.2	48.1	58.4	58.5	66.2	64.9	54.1	26.1	43.9	55.9	69.8	50.9	53.0
ImageNet-R-CNN[19]	64.2	69.7	50	41.9	32.0	62.6	71.0	60.7	32.7	58.5	46.5	56.1	60.6	66.8	54.2	31.5	52.8	48.9	57.9	64.7	54.2

Table 1. Results on VOC-2007. R-CNN performance with our unsupervised pre-training is 5% MAP better than training from scratch, but still 8% below pre-training with ImageNet label supervision.

Doersch et al., ICCV 2015, Unsupervised visual representation learning by context prediction

Pretext Task2: Inpainting

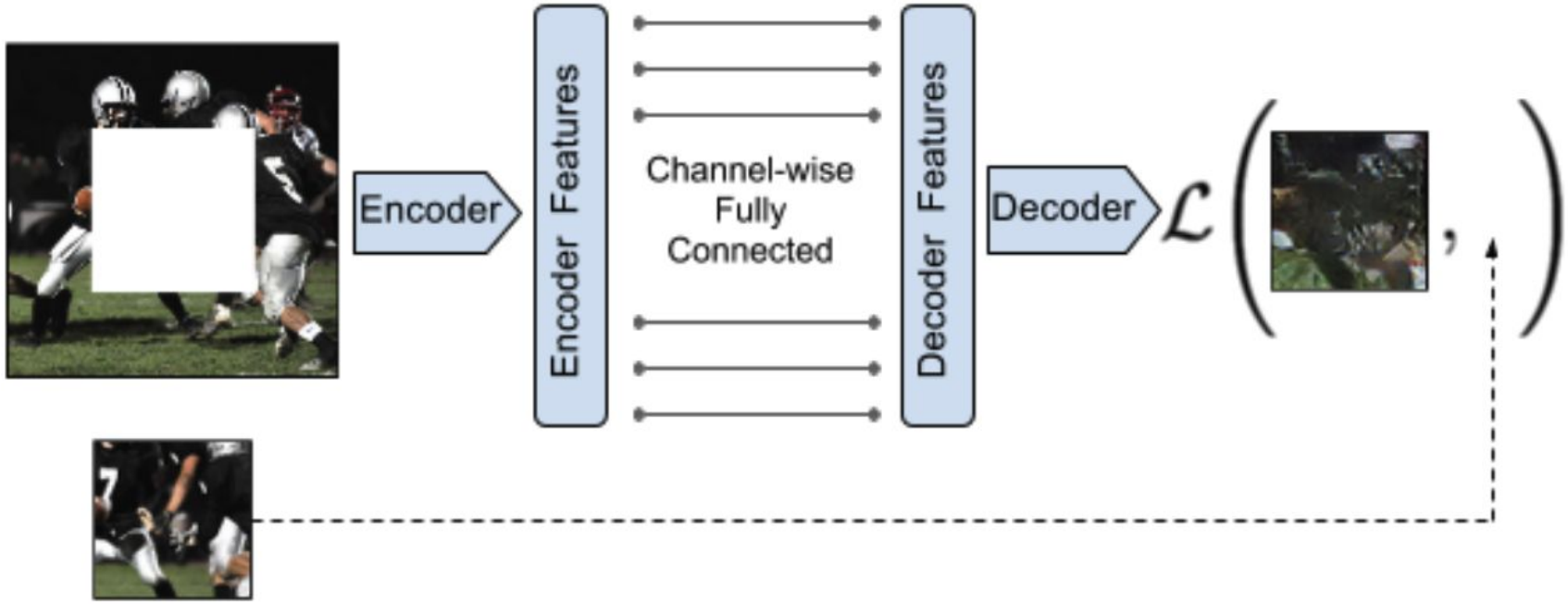
- Can we predict missing regions from surrounding context?
- To fill the missing region, the network must understand:
  - Object parts
  - Scene structure
  - Semantic context



Pathak et al., CVPR 2016, Context Encoders: Feature Learning by Inpainting

Pretext Task2: Inpainting

- Learn from reconstruction.

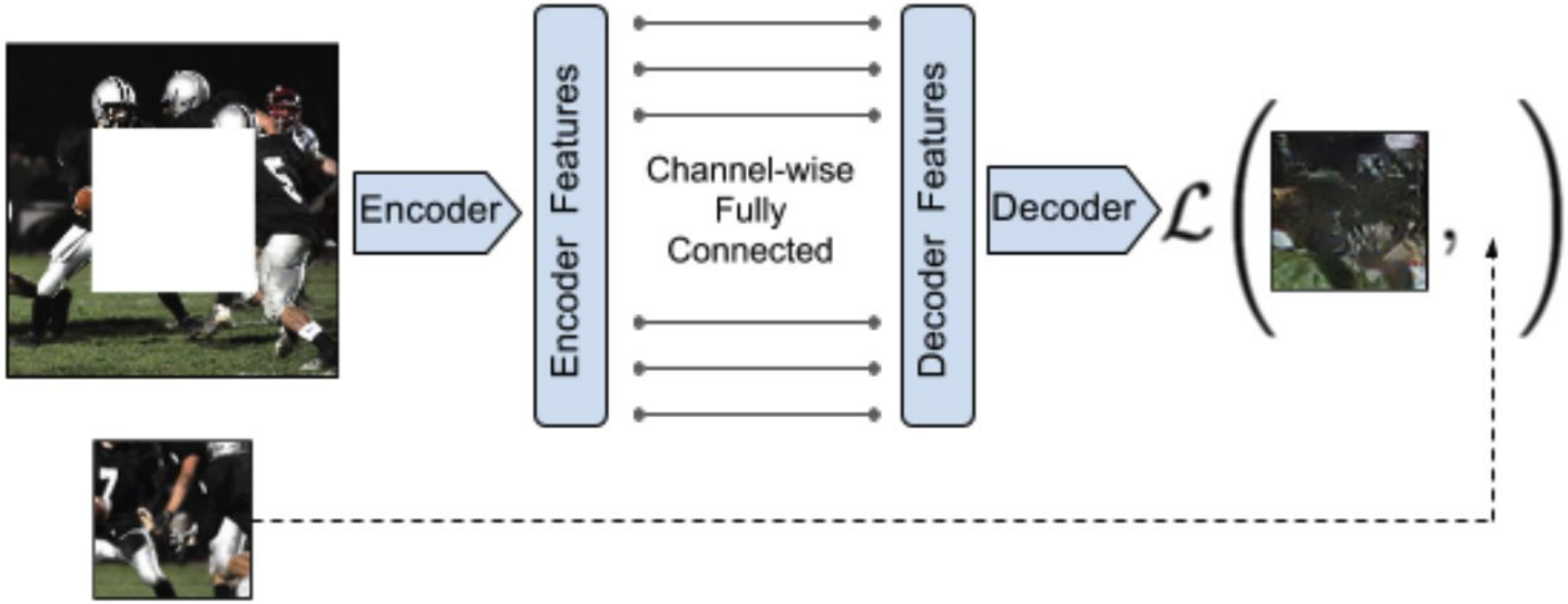


$$\mathcal{L}_{rec}(x) = \|\hat{M} \odot (x - F((1 - \hat{M}) \odot x))\|_2^2$$

Pathak et al., CVPR 2016, Context Encoders: Feature Learning by Inpainting

Pretext Task2: Inpainting

- Learn from reconstruction + adversarial loss.

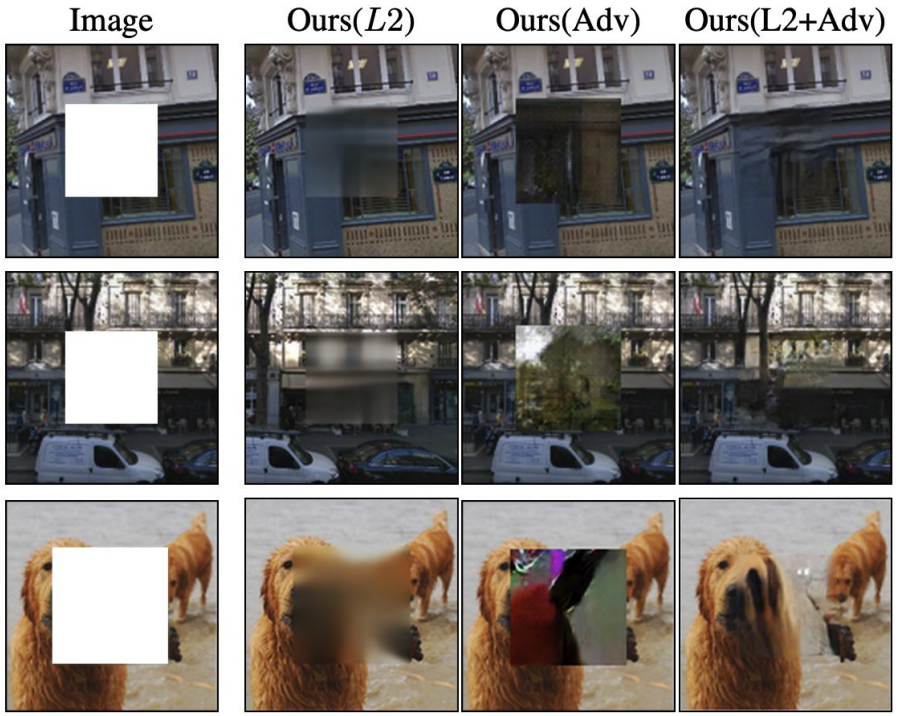


$$\mathcal{L}_{rec}(x) = \|\hat{M} \odot (x - F((1 - \hat{M}) \odot x))\|_2^2$$

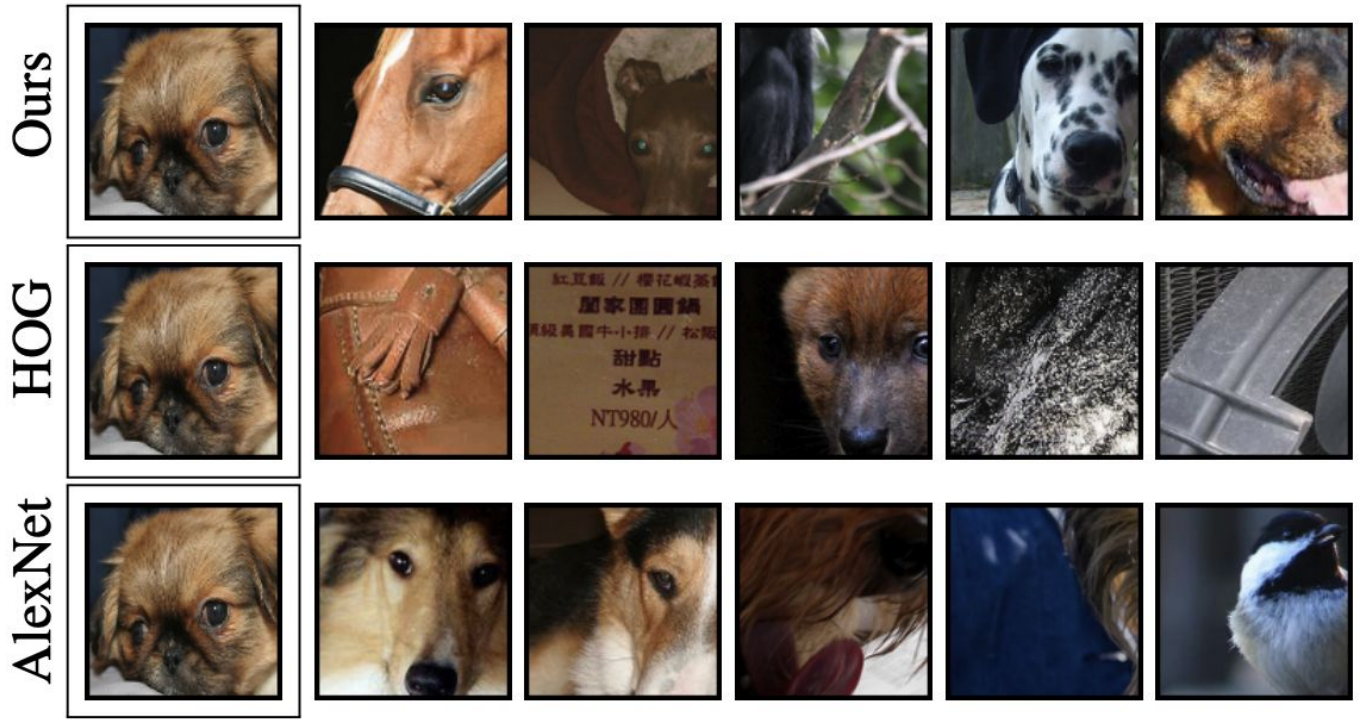
$$\mathcal{L}_{adv} = \max_D \quad \mathbb{E}_{x \in \mathcal{X}}[\log(D(x)) + \log(1 - D(F((1 - \hat{M}) \odot x)))]$$

Pathak et al., CVPR 2016, Context Encoders: Feature Learning by Inpainting

Pretext Task2: Inpainting



Qualitative Results



KNN Results

Pathak et al., CVPR 2016, Context Encoders: Feature Learning by Inpainting

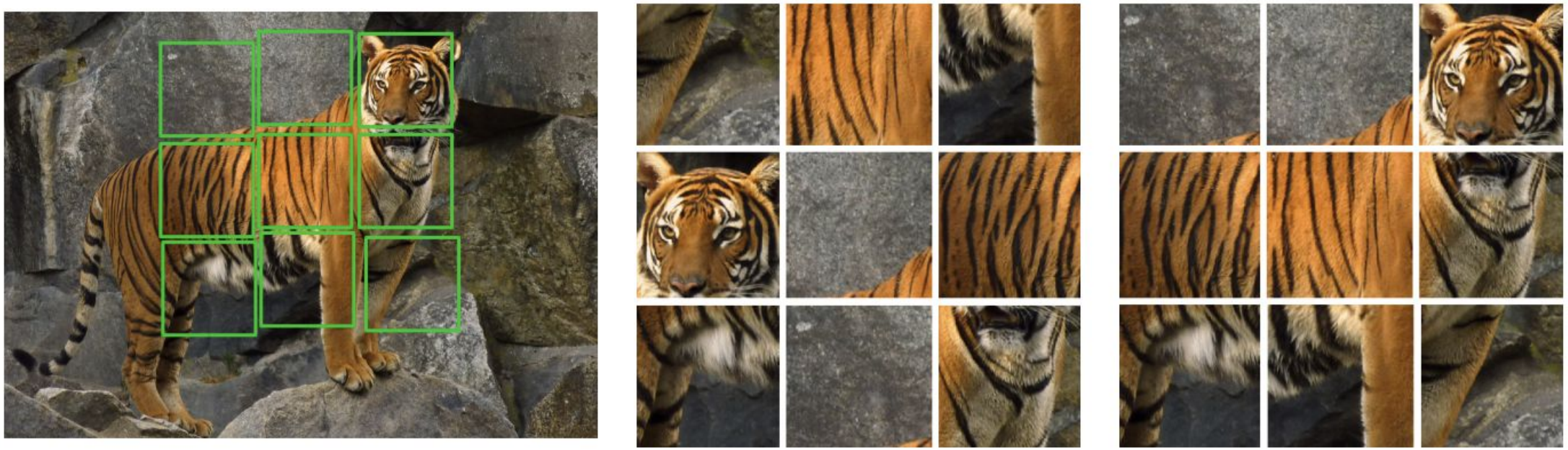
Pretext Task2: Inpainting

Pretraining Method	Supervision	Pretraining time	Classification	Detection	Segmentation
ImageNet [26]	1000 class labels	3 days	78.2%	56.8%	48.0%
Random Gaussian	initialization	< 1 minute	53.3%	43.4%	19.8%
Autoencoder	-	14 hours	53.8%	41.9%	25.2%
Agrawal *et al.* [1]	egomotion	10 hours	52.9%	41.8%	-
Wang *et al.* [39]	motion	1 week	58.7%	47.4%	-
Doersch *et al.* [7]	relative context	4 weeks	55.3%	46.6%	-
Ours	context	14 hours	56.5%	44.5%	30.0%

Relative Position of Patches

Pathak et al., CVPR 2016, Context Encoders: Feature Learning by Inpainting

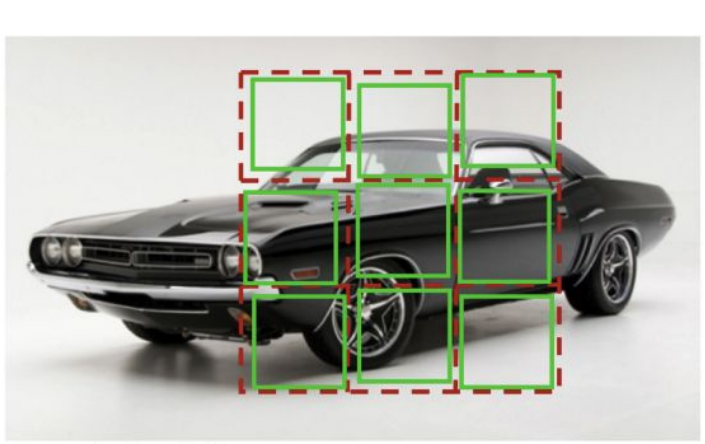
Pretext Task3: Jigsaw Puzzles



How can we solve this Jigsaw Puzzles?

Noroozi et al., ECCV 2016, Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles

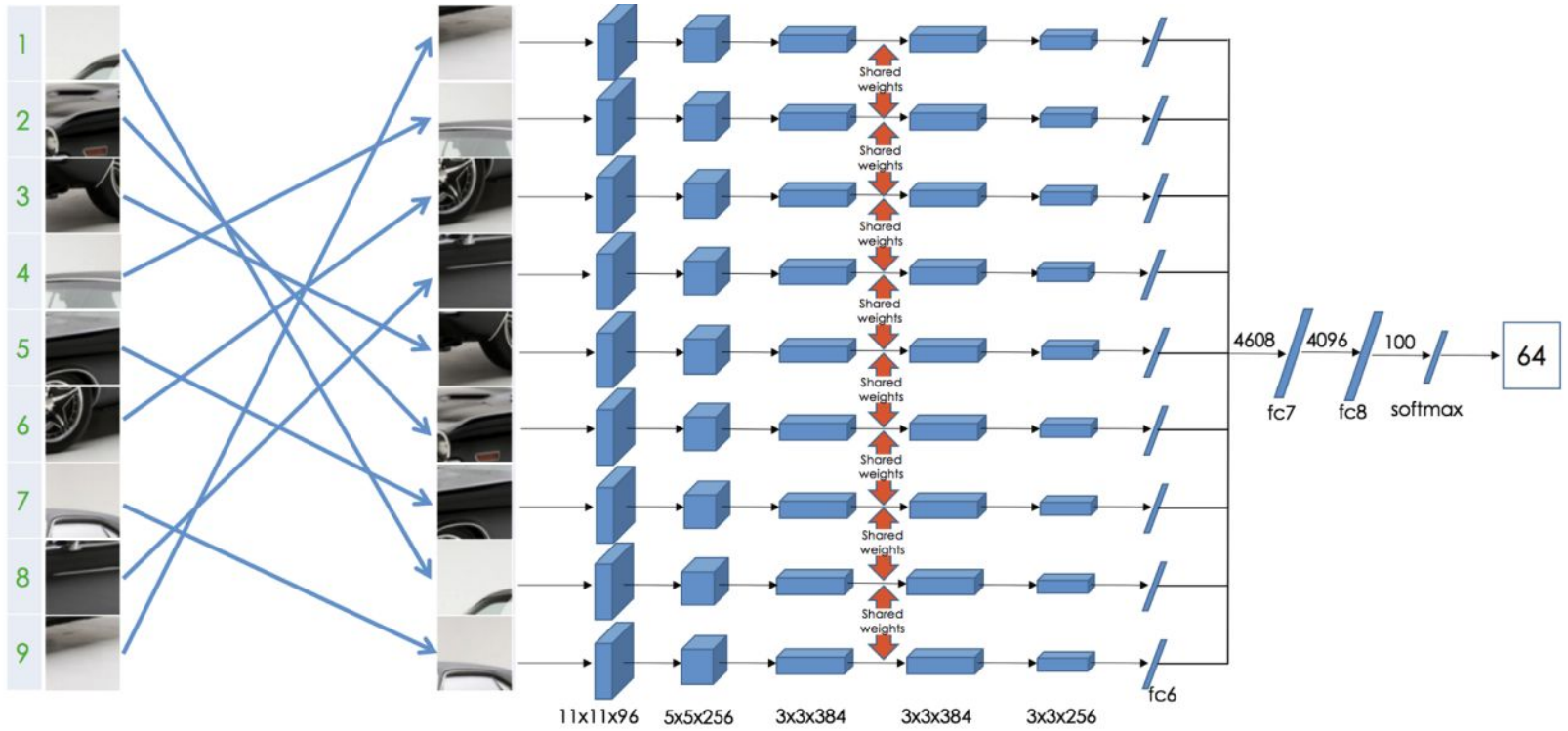
Pretext Task3: Jigsaw Puzzles



Permutation Set

index	permutation
64	9,4,6,8,3,2,5,1,7

Reorder patches according to the selected permutation



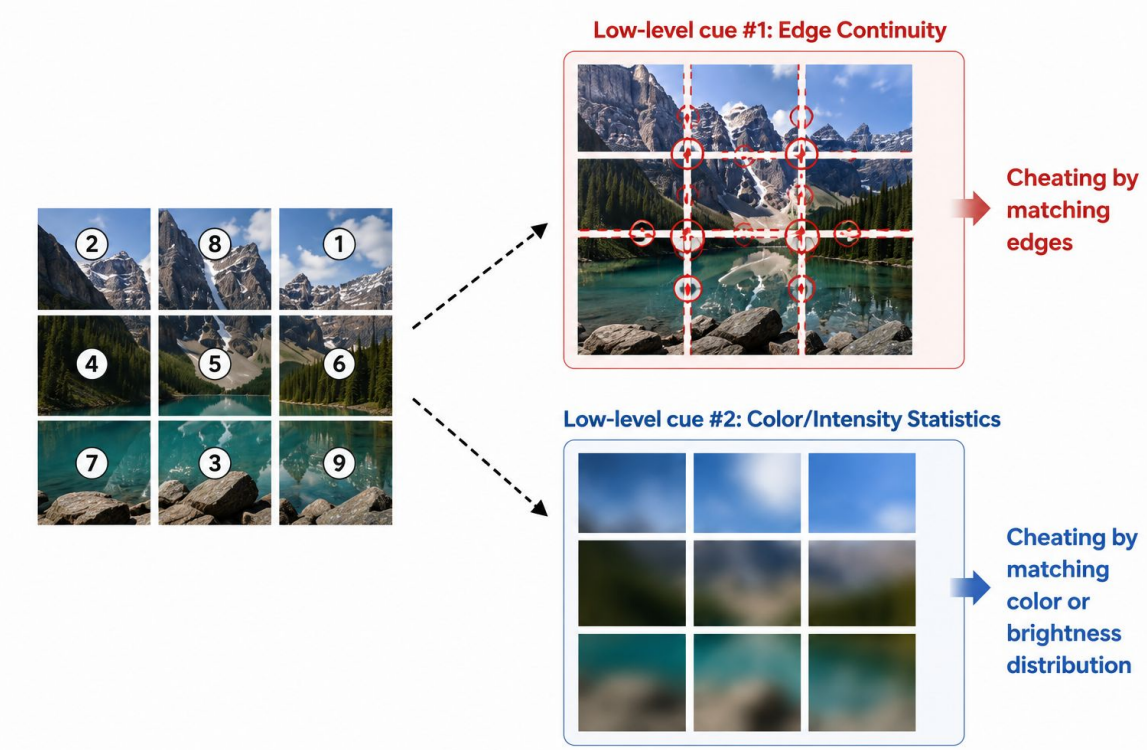
Given 9 tiles, there are 9! = 362,880 possible permutations

- Too many classes for efficient training
- Select a subset of 64 permutations that maximize pairwise Hamming distance between permutations

Noroozi et al., ECCV 2016, Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles

Pretext Task3: Jigsaw Puzzles

- Preventing Cheating Problem:
  - The network may solve the puzzle using low-level cues instead of semantics.
  - Solutions:
    - Multiple puzzle configurations per image
    - Random gaps between patches
    - Random crop
    - Spatial jitter
    - Color jitter
    - Grayscale images



Noroozi et al., ECCV 2016, Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles

Pretext Task3: Jigsaw Puzzles

Table 1: Results on PASCAL VOC 2007 Detection and Classification. The results of the other methods are taken from Pathak *et al.* [30].

Method	Pretraining time	Supervision	Classification	Detection	Segmentation
Krizhevsky *et al.* [25]	3 days	1000 class labels	**78.2%**	**56.8%**	**48.0%**
Wang and Gupta[39]	1 week	motion	58.4%	44.0%	-
Doersch *et al.* [10]	4 weeks	context	55.3%	46.6%	-
Pathak *et al.* [30]	14 hours	context	56.5%	44.5%	29.7%
Ours	2.5 days	context	**67.6%**	**53.2%**	**37.6%**

Relative Position of Patches

Inpainting

Noroozi et al., ECCV 2016, Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles

Pretext Task3: Jigsaw Puzzles

Table 4: Ablation study on the impact of the permutation set.

Number of permutations	Average hamming distance	Minimum hamming distance	Jigsaw task accuracy	Detection performance
1000	8.00	2	71	**53.2**
1000	6.35	2	62	51.3
1000	3.99	2	54	50.2
100	8.08	2	88	52.6
95	8.08	3	90	52.4
85	8.07	4	91	52.7
71	8.07	5	92	52.8
35	8.13	6	94	52.6
10	8.57	7	97	49.2
7	8.95	8	98	49.6
6	9	9	99	49.7

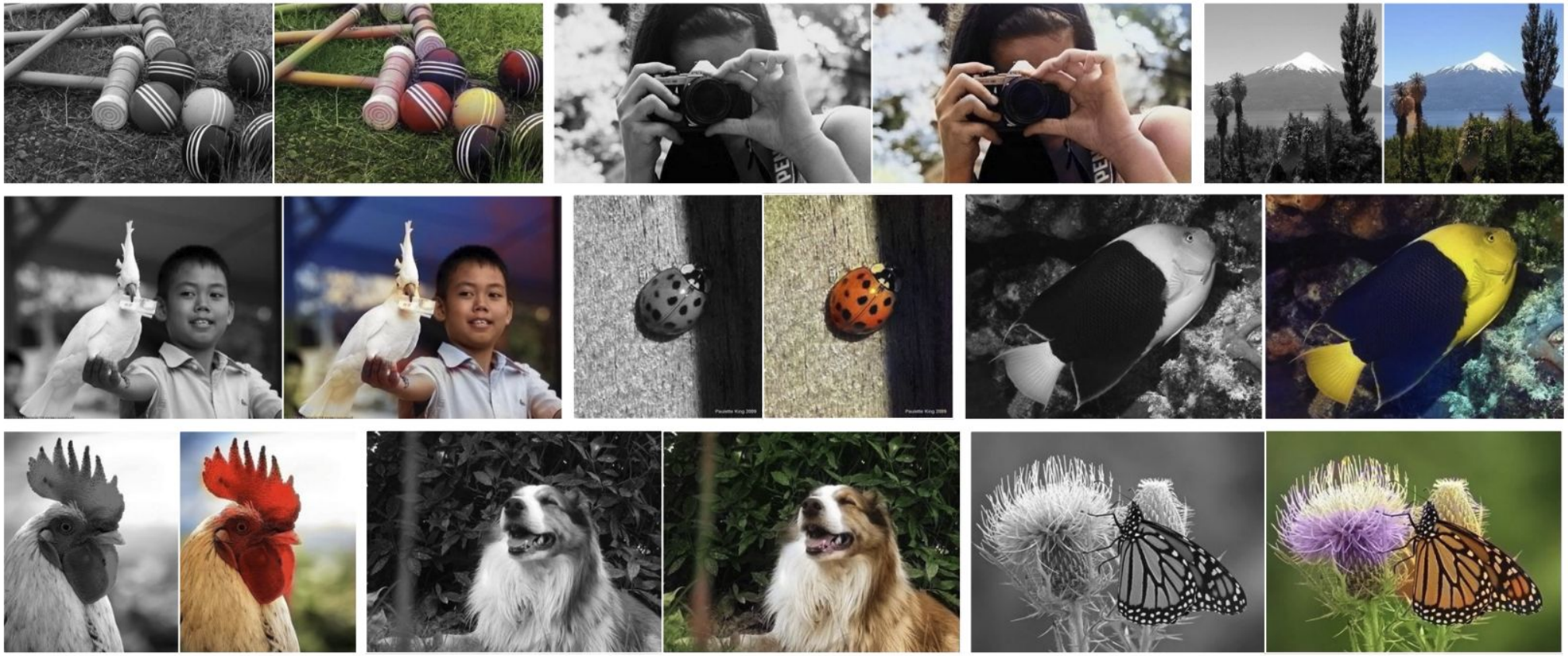
Table 5: Ablation study on the impact of the shortcuts.

Gap	Normalization	Color jittering	Jigsaw task accuracy	Detection performance
✗	✓	✓	98	47.7
✓	✗	✓	90	43.5
✓	✓	✗	89	51.1
✓	✓	✓	88	52.6

Noroozi et al., ECCV 2016, Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles

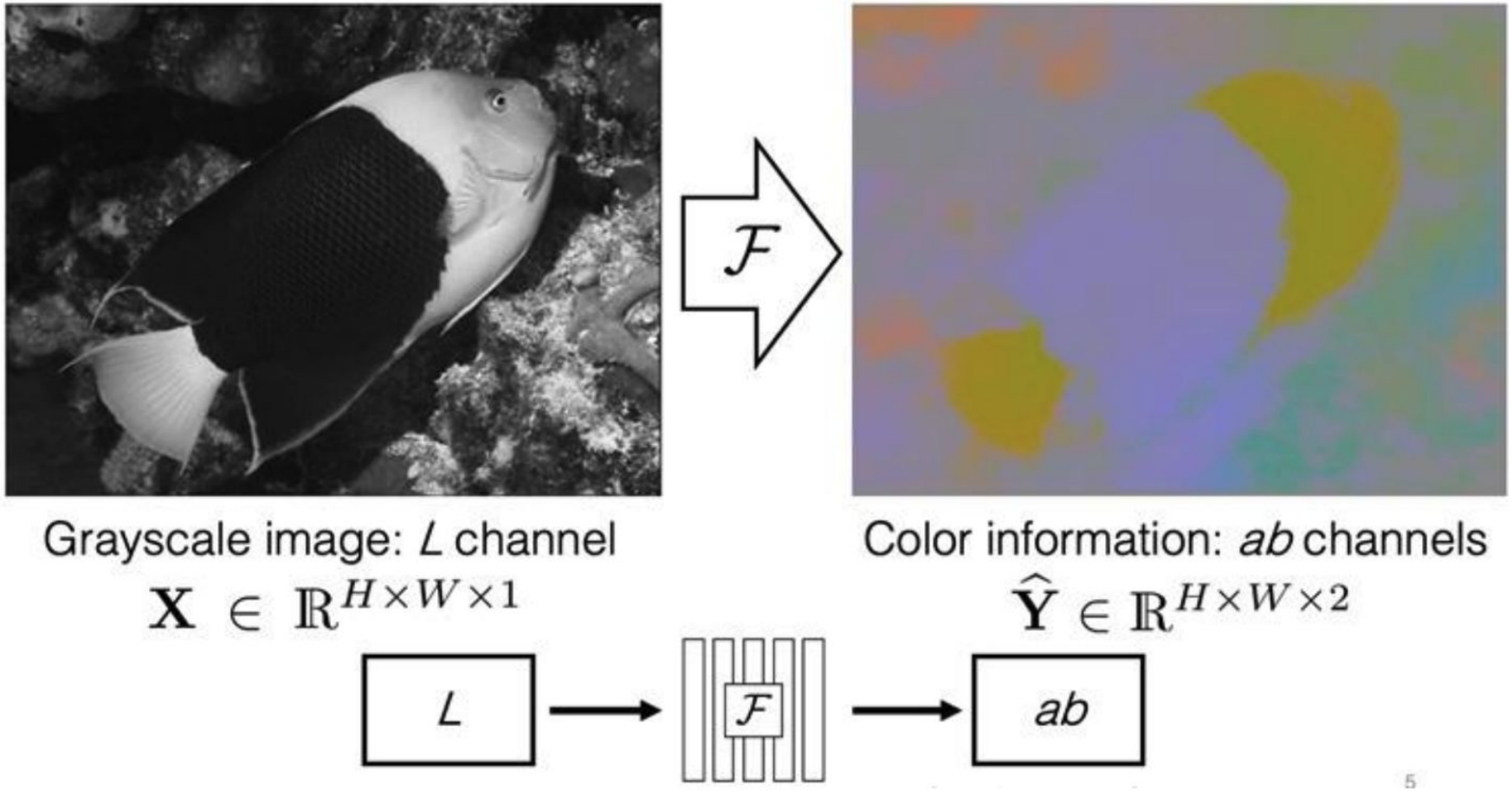
Pretext Task4: Colorization

- Can a network recover colors from grayscale images?
  - Color prediction requires semantic understanding.



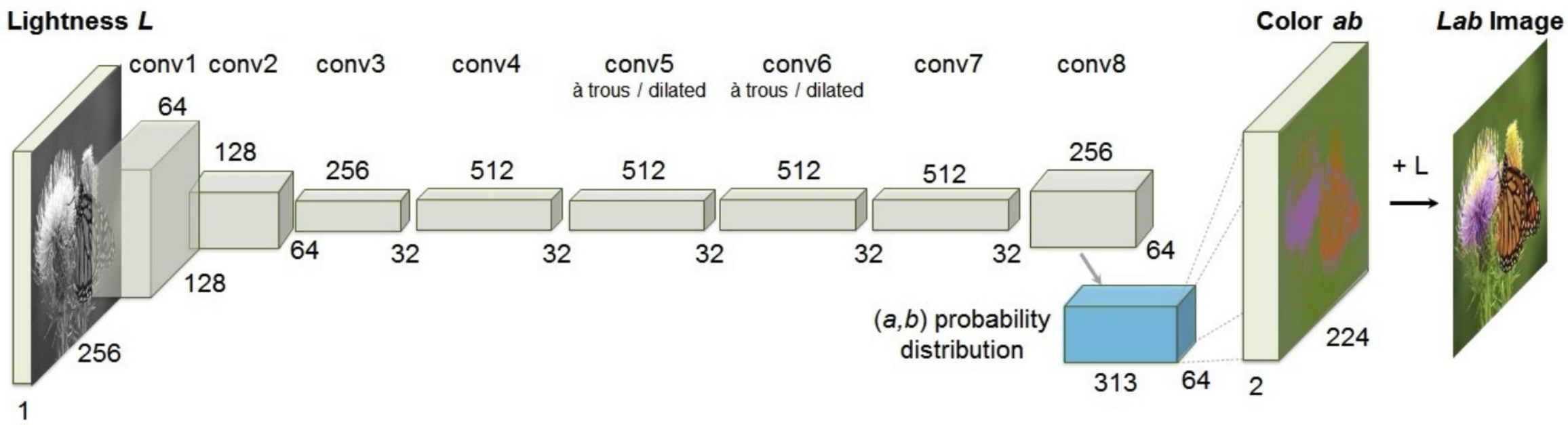
Zhang et al., ECCV 2016, Colorful Image Colorization

Pretext Task4: Colorization



Zhang et al., ECCV 2016, Colorful Image Colorization

Pretext Task4: Colorization



$$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$$

$$\hat{\mathbf{Y}} = \mathcal{F}(\mathbf{X})$$

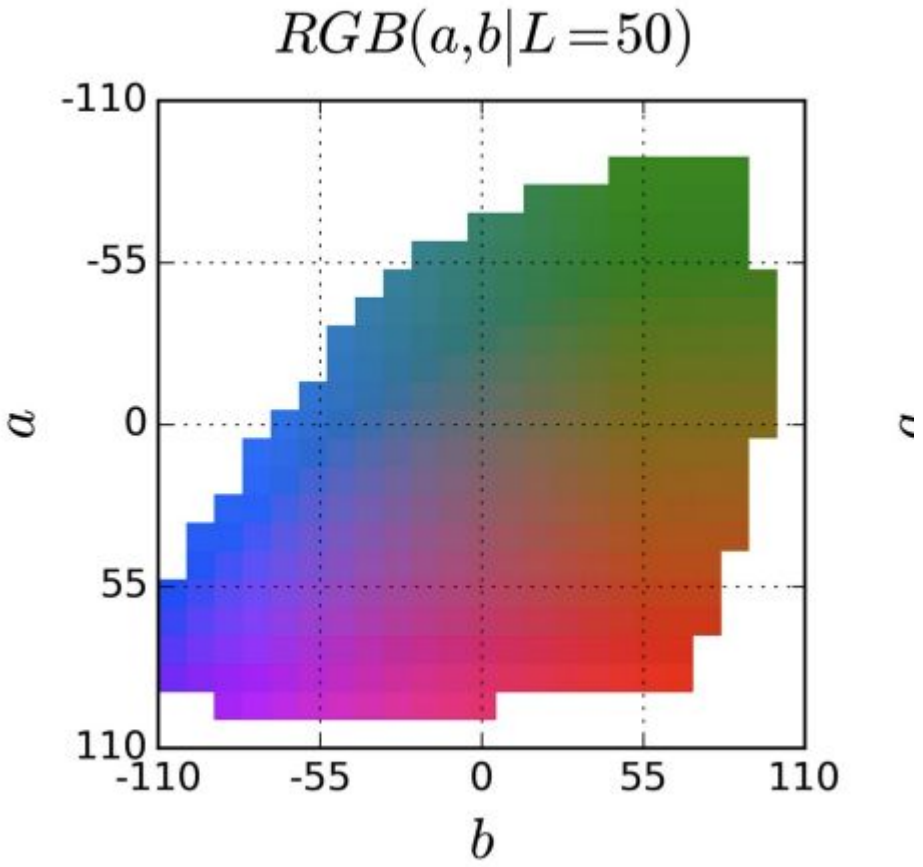
$$\mathrm{L}_2(\hat{\mathbf{Y}}, \mathbf{Y}) = \frac{1}{2} \sum_{h,w} \|\mathbf{Y}_{h,w} - \hat{\mathbf{Y}}_{h,w}\|_2^2$$

Zhang et al., ECCV 2016, Colorful Image Colorization

Pretext Task4: Colorization

- Can it be solved by classification instead of regression.
  - Multiple colors can be plausible.
  - Regression averages them.
  - Continuous color prediction
  - → 313 color classes

$$\mathrm{L}_{cl}(\hat{\mathbf{Z}}, \mathbf{Z}) = -\sum_{h,w} v(\mathbf{Z}_{h,w}) \sum_{q} \mathbf{Z}_{h,w,q} \log(\hat{\mathbf{Z}}_{h,w,q})$$



Zhang et al., ECCV 2016, Colorful Image Colorization

Pretext Task4: Colorization

- Qualitative Results



Regression Classification

Zhang et al., ECCV 2016, Colorful Image Colorization

Pretext Task4: Colorization

Dataset and Task Generalization on PASCAL [37]								
		Class. (%mAP)			Det. (%mAP)		Seg. (%mIU)	
fine-tune layers	**[Ref]**	**fc8**	**fc6-8**	**all**	**[Ref]**	**all**	**[Ref]**	**all**
ImageNet [38]	-	76.8	78.9	79.9	[36]	56.8	[42]	48.0
Gaussian	[10]	–	–	53.3	[10]	43.4	[10]	19.8
Autoencoder	[16]	24.8	16.0	53.8	[10]	41.9	[10]	25.2
k-means [36]	[16]	32.0	39.2	56.6	[36]	45.6	[16]	32.6
Agrawal et al. [8]	[16]	31.2	31.0	54.2	[36]	43.9	–	–
Wang & Gupta [15]	–	28.1	52.2	58.7	[36]	47.4	–	–
*Doersch et al. [14]	[16]	44.7	55.1	**65.3**	[36]	**51.1**	–	–
*Pathak et al. [10]	[10]	–	–	56.5	[10]	44.5	[10]	29.7
*Donahue et al. [16]	–	38.2	50.2	58.6	[16]	46.2	[16]	34.9
Ours (gray)	–	**52.4**	**61.5**	**65.9**	–	46.1	–	35.0
Ours (color)	–	**52.4**	**61.5**	**65.6**	–	46.9	–	**35.6**

Zhang et al., ECCV 2016, Colorful Image Colorization

Pretext Task5: Rotation Prediction

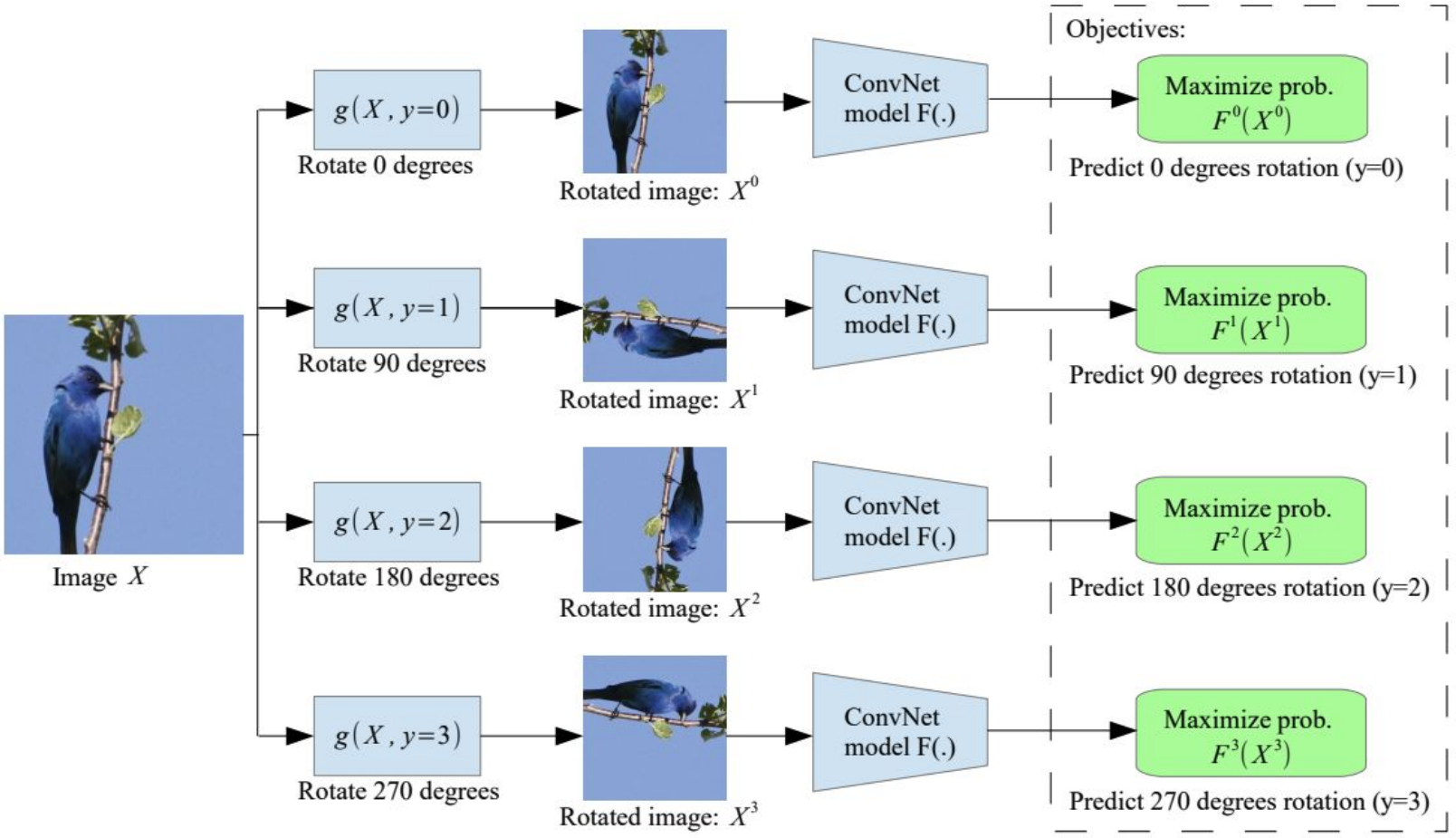
- Which orientation is correct?
  - To recognize rotation, the network must recognize objects.



Gidaris et al., ICLR 2018, UNSUPERVISED REPRESENTATION LEARNING BY PREDICTING IMAGE ROTATIONS

Pretext Task5: Rotation Prediction

- No labels are needed, The applied rotation becomes the label.



Gidaris et al., ICLR 2018, UNSUPERVISED REPRESENTATION LEARNING BY PREDICTING IMAGE ROTATIONS

Pretext Task5: Rotation Prediction

- Rotation prediction accuracy.

Table 2: Exploring the quality of the self-supervised learned features w.r.t. the number of recognized rotations. For all the entries we trained a non-linear classifier with 3 fully connected layers (similar to Table 1) on top of the feature maps generated by the 2nd conv. block of a RotNet model with 4 conv. blocks in total. The reported results are from CIFAR-10.

# Rotations	Rotations	CIFAR-10 Classification Accuracy
4	0°, 90°, 180°, 270°	**89.06**
8	0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°	88.51
2	0°, 180°	87.46
2	90°, 270°	85.52

Gidaris et al., ICLR 2018, UNSUPERVISED REPRESENTATION LEARNING BY PREDICTING IMAGE ROTATIONS

Pretext Task5: Rotation Prediction

- Downstream results.

Inpainting

	Classification (%mAP)		Detection (%mAP)	Segmentation (%mIoU)
Trained layers	fc6-8	all	all	all
ImageNet labels	78.9	79.9	56.8	48.0
Random		53.3	43.4	19.8
Random rescaled Krähenbühl et al. (2015)	39.2	56.6	45.6	32.6
Egomotion (Agrawal et al., 2015)	31.0	54.2	43.9	
Context Encoders (Pathak et al., 2016b)	34.6	56.5	44.5	29.7
Tracking (Wang & Gupta, 2015)	55.6	63.1	47.4	
Context (Doersch et al., 2015)	55.1	65.3	51.1	
Colorization (Zhang et al., 2016a)	61.5	65.6	46.9	35.6
BIGAN (Donahue et al., 2016)	52.3	60.1	46.9	34.9
Jigsaw Puzzles (Noroozi & Favaro, 2016)	-	67.6	53.2	37.6
NAT (Bojanowski & Joulin, 2017)	56.7	65.3	49.4	
Split-Brain (Zhang et al., 2016b)	63.0	67.1	46.7	36.0
ColorProxy (Larsson et al., 2017)		65.9		38.4
Counting (Noroozi et al., 2017)	-	67.7	51.4	36.6
(Ours) RotNet	**70.87**	**72.97**	**54.4**	**39.1**

Gidaris et al., ICLR 2018, UNSUPERVISED REPRESENTATION LEARNING BY PREDICTING IMAGE ROTATIONS

Summary of Pretext Tasks

Pretext Tasks

Task	Core Question	Learned Concept
Relative Position	Where is the neighboring patch?	Local Spatial Context
Jigsaw	How should patches be arranged?	Global Spatial Structure
Inpainting	What is missing?	Context & Object Parts
Colorization	What color should it be?	Object Semantics
Rotation	Which orientation is correct?	Object-Centric Understanding

Pretext Tasks

- Pretext tasks focus on "**visual common sense**", e.g., predict rotations, inpainting, rearrangement, and colorization.
- The models are forced learn good features about natural images, e.g., semantic representation of an object category, in order to solve the pretext tasks.
- We often do not care about the performance of these pretext tasks, but rather how useful the learned features are for downstream tasks (classification, detection, segmentation).

Problems: 1) coming up with individual pretext tasks is tedious.

The Hope of Generalization

- We really hope that the pre-training task and the transfer task are "aligned".



#sun #nofilter #fun
#tree #aruba

Pre-training

Weak or self-supervised

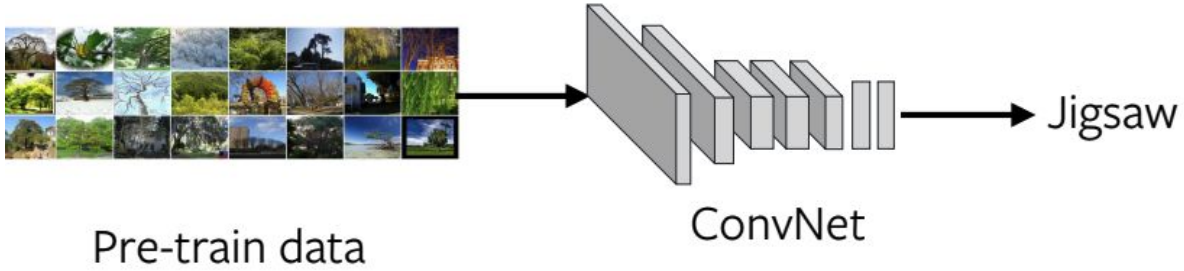


Transfer Tasks

Why should solving Jigsaw puzzles teach about "semantics"?

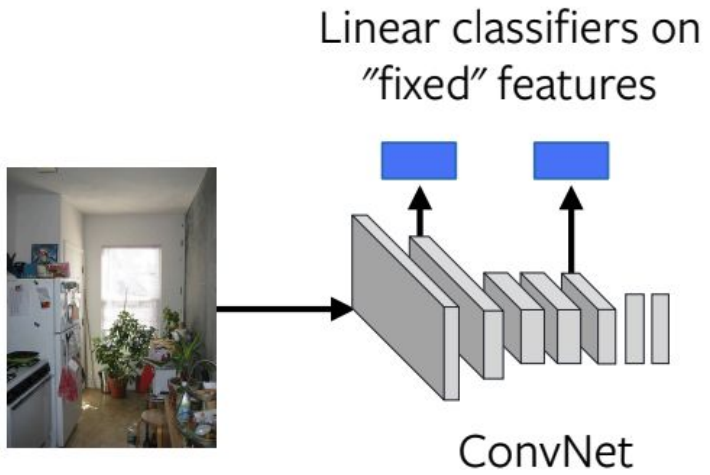
Why should performing a non semantic task produce good features?

The Hope of Generalization



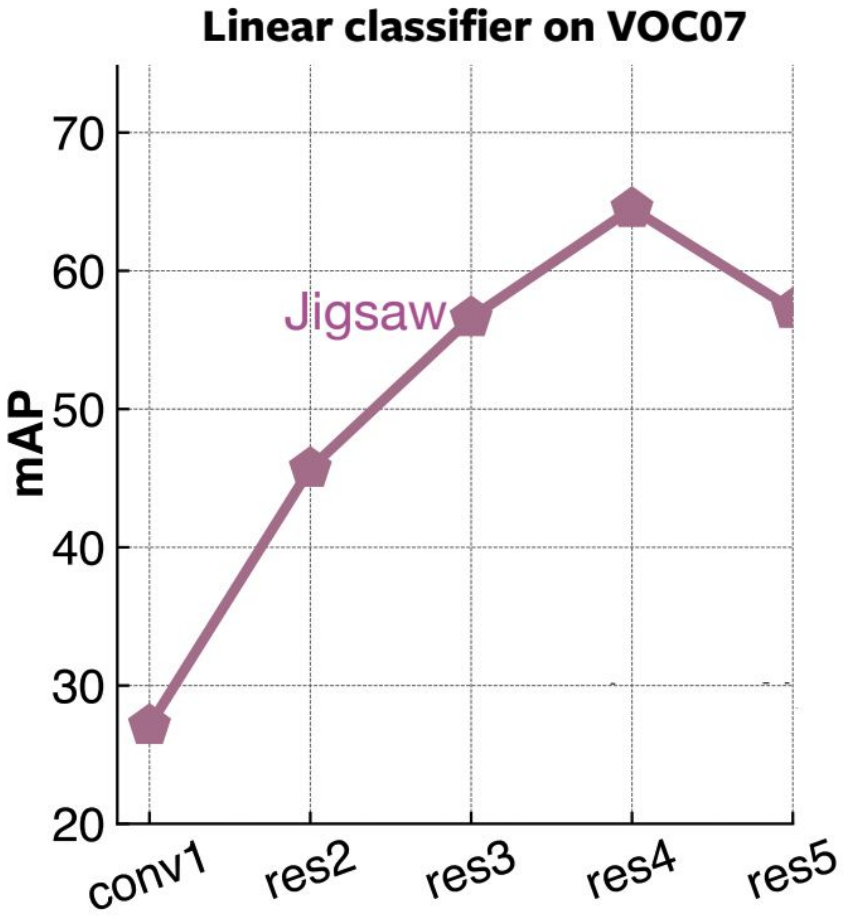
Pre-training

Weak or self-supervised

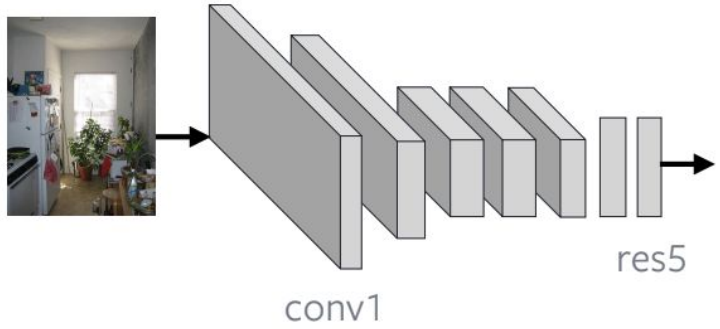


Transfer

Higher layers do not generalize ...



mAP = mean Average Precision (Higher is better)



Problems: 2) the learned representations may not be general.

Multiple Pretext Tasks



Jigsaw Puzzle
Inpainting
Colorization
…

Kim et al., WACV 2018, Learning Image Representations by Completing Damaged Jigsaw Puzzles

Multiple Pretext Tasks

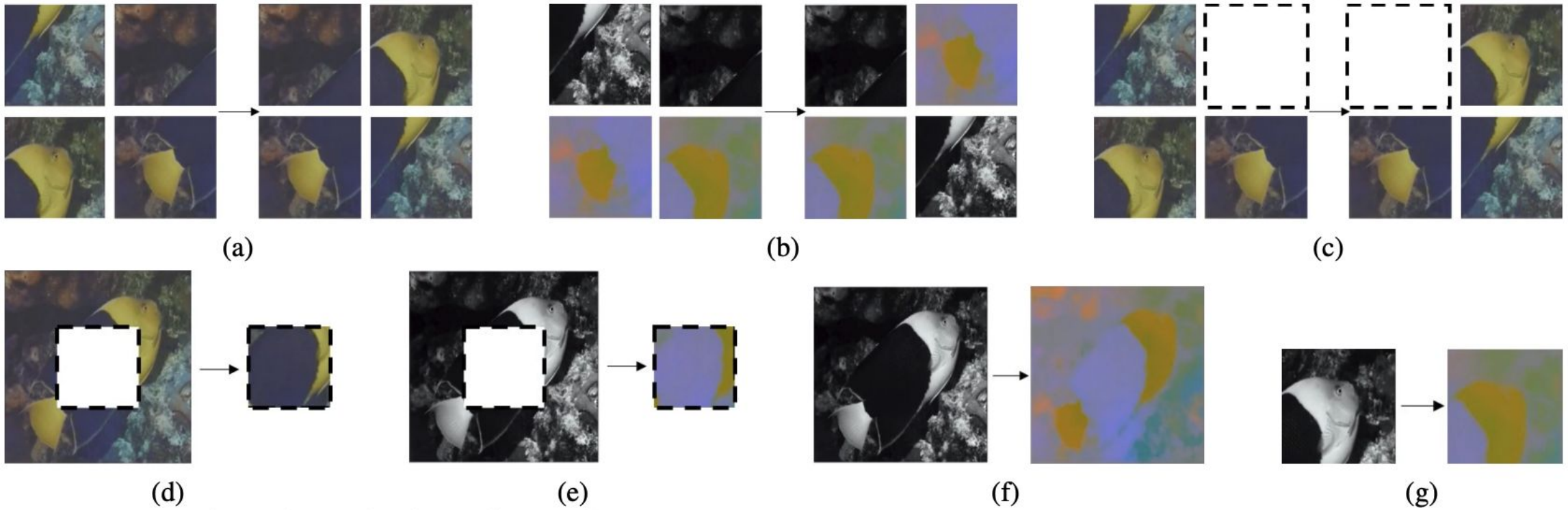
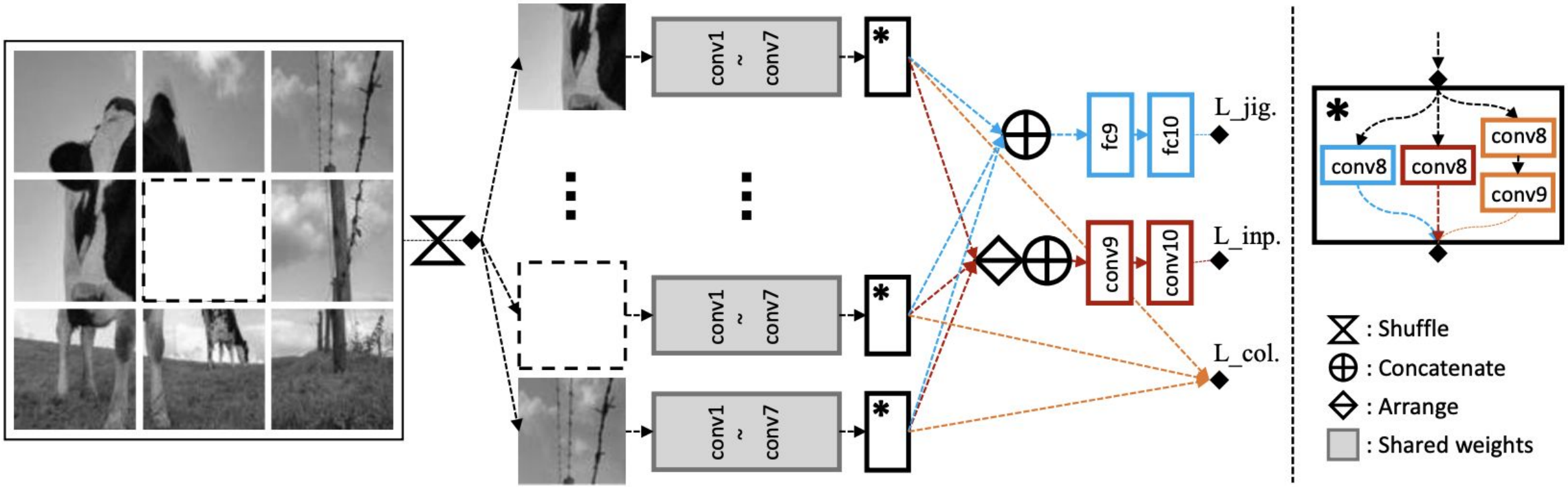


Figure 2. **Illustrations of complicating self-supervised tasks.** (a) Conventional 2×2 jigsaw puzzles. (b) Complicated 2×2 jigsaw puzzles; each patch's L or ab channel is dropped. (c) Complicated 2×2 jigsaw puzzles; one of the patches is completely dropped. (d) Conventional inpainting. (e) Complicated inpainting; it outputs in ab channels from an input in only L channel. (f) Conventional colorization. (g) Complicated colorization; only one-quarter of the entire image is given for colorization.

Kim et al., WACV 2018, Learning Image Representations by Completing Damaged Jigsaw Puzzles

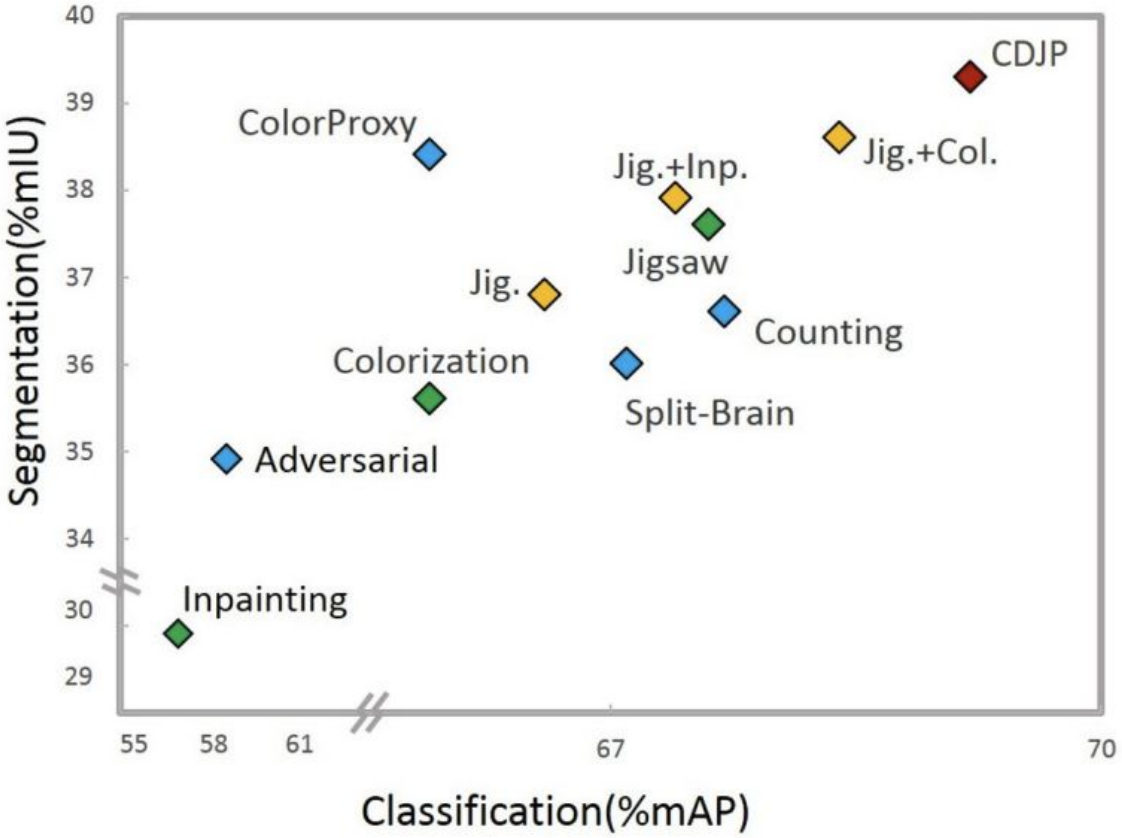
Multiple Pretext Tasks



Kim et al., WACV 2018, Learning Image Representations by Completing Damaged Jigsaw Puzzles

Multiple Pretext Tasks

Method	Class.	Det.	Segm.
ImageNet [20]	79.9	56.8	48.0
Random	53.3	43.4	19.8
RelativePosition [4]	65.3	51.1	-
Jigsaw [25]	67.6	**53.2**	37.6
Ego-motion [36]	54.2	43.9	-
Adversarial [6]	58.6	46.2	34.9
Inpainting [30]	56.5	44.5	29.7
Colorization [38]	65.9	46.9	35.6
Split-Brain [39]	67.1	46.7	36.0
ColorProxy [21]	65.9	-	38.4
WatchingObjectMove [29]	61.0	52.2	-
Counting [26]	67.7	51.4	36.6
CDJP	**69.2**	52.4	**39.3**



Kim et al., WACV 2018, Learning Image Representations by Completing Damaged Jigsaw Puzzles

Toward Contrastive Learning

Pretext tasks from image transformations

- Learned representations may be tied to a specific pretext task (even multi tasks)!
- Can we come up with a more general pretext task?

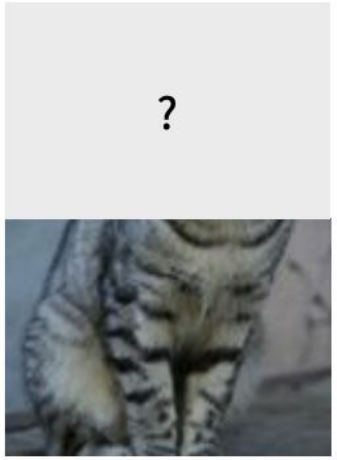
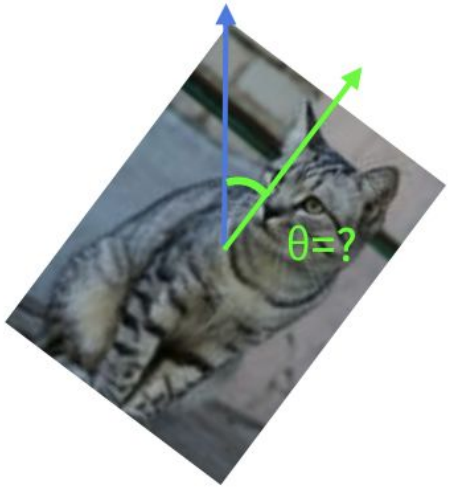


image completion



rotation prediction

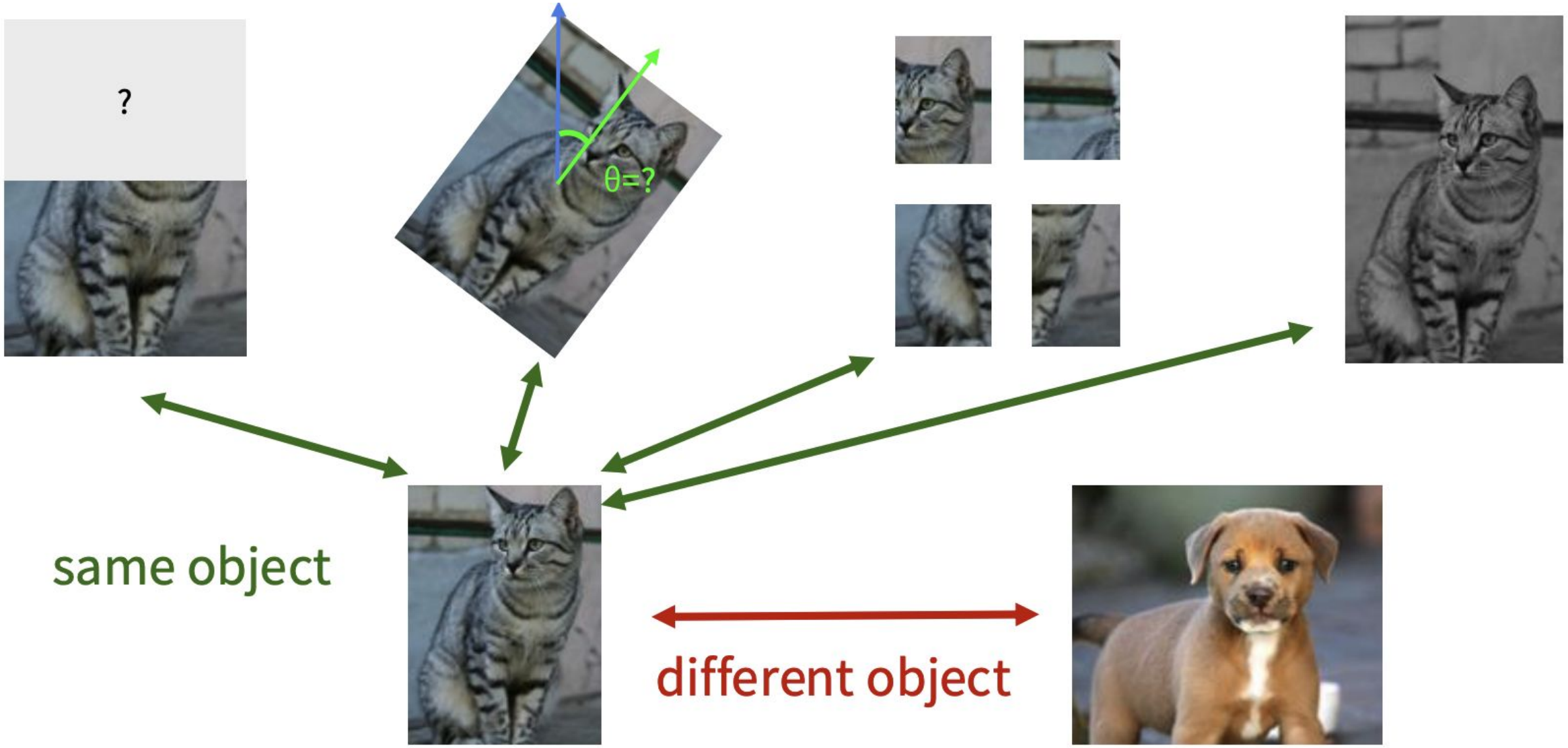


“jigsaw puzzle”



colorization

A more general pretext task?



The Thirty-Third AAAI Conference on Artificial Intelligence (AAAI-19)

Self-Supervised Video Representation Learning with Space-Time Cubic Puzzles

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Kim et al., AAAI 2019, Self-Supervised Video Representation Learning with Space-Time Cubic Puzzles

From Image Representation to Video Representation

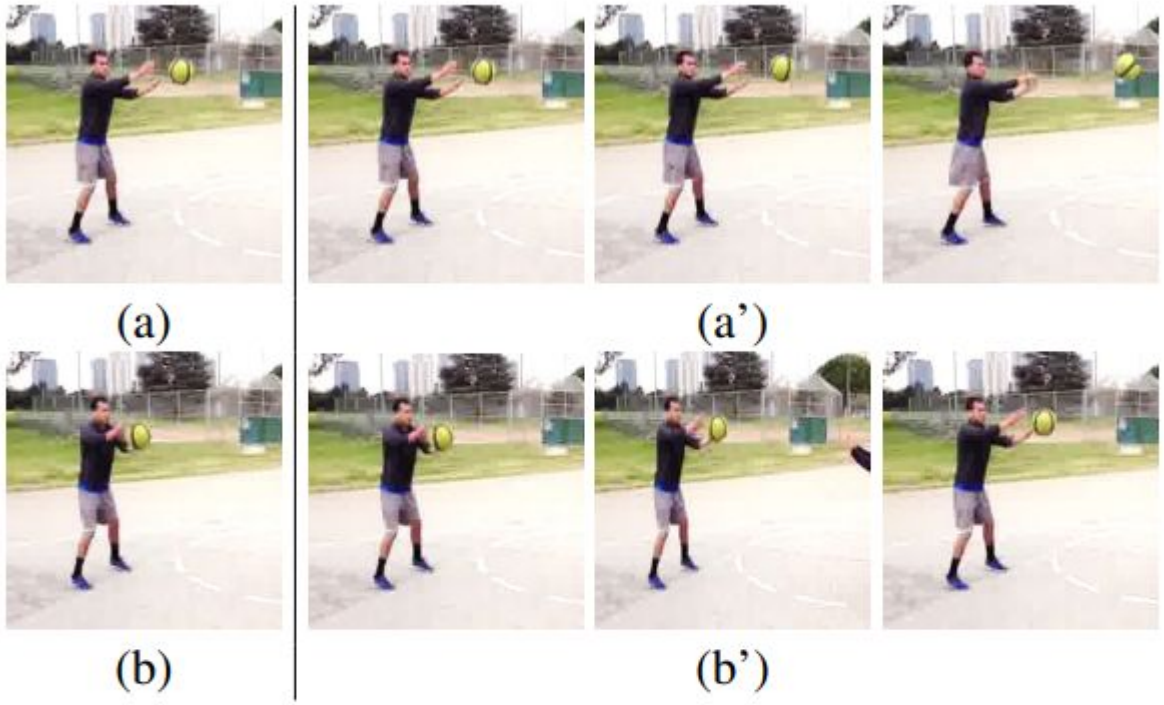


Figure 1: **Ambiguity in time direction.** Given a pair of images on the left, it is ambiguous to determine the time direction in between. One can finally identify the direction (b' → a') and action (*throwing*) when given video sequences.

An image provides spatial appearance.

A video additionally provides:
- Temporal order
- Motion
- Changes over time

Can we understand an action from two frames?

Space-Time Cubic Puzzle

Pretext task:

- Arrange shuffled video cubes
- into their original space-time order.

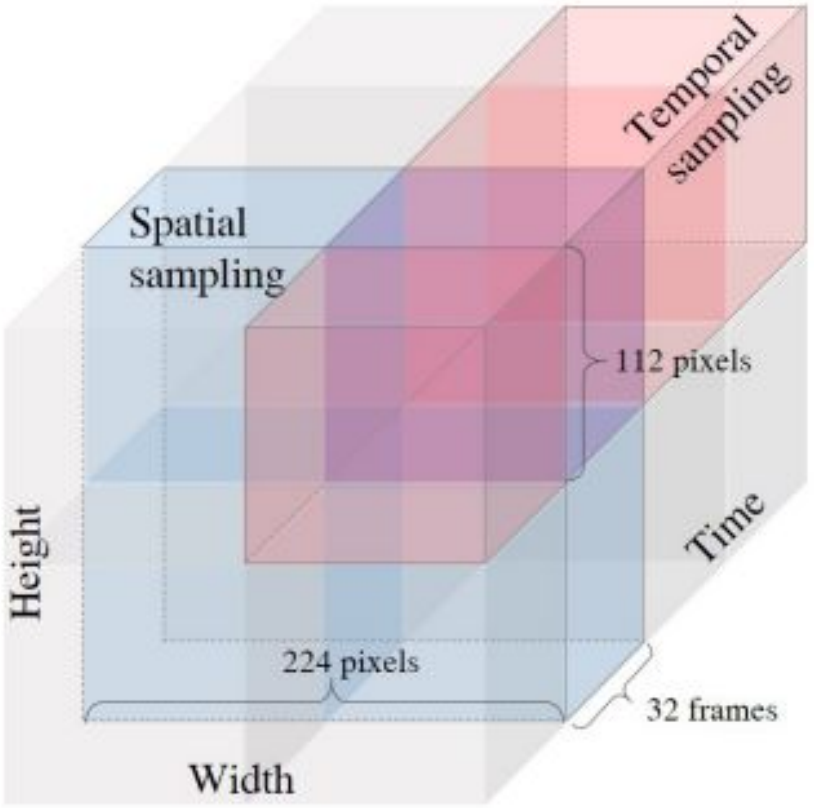
Spatial Puzzle

- Where was each cube located?

Temporal Puzzle

- When did each cube occur?

Generating Space-Time Cubic Puzzles



1. Divide a video into a 2 × 2 × 4 grid.

2. Sample four video cubes.

   Spatial: 2 × 2 × 1
   Temporal: 1 × 1 × 4

3. Randomly permute the cubes.
4. Predict the original arrangement.

→ 4 cubes produce 4! = 24 possible permutations.

Generating Space-Time Cubic Puzzles

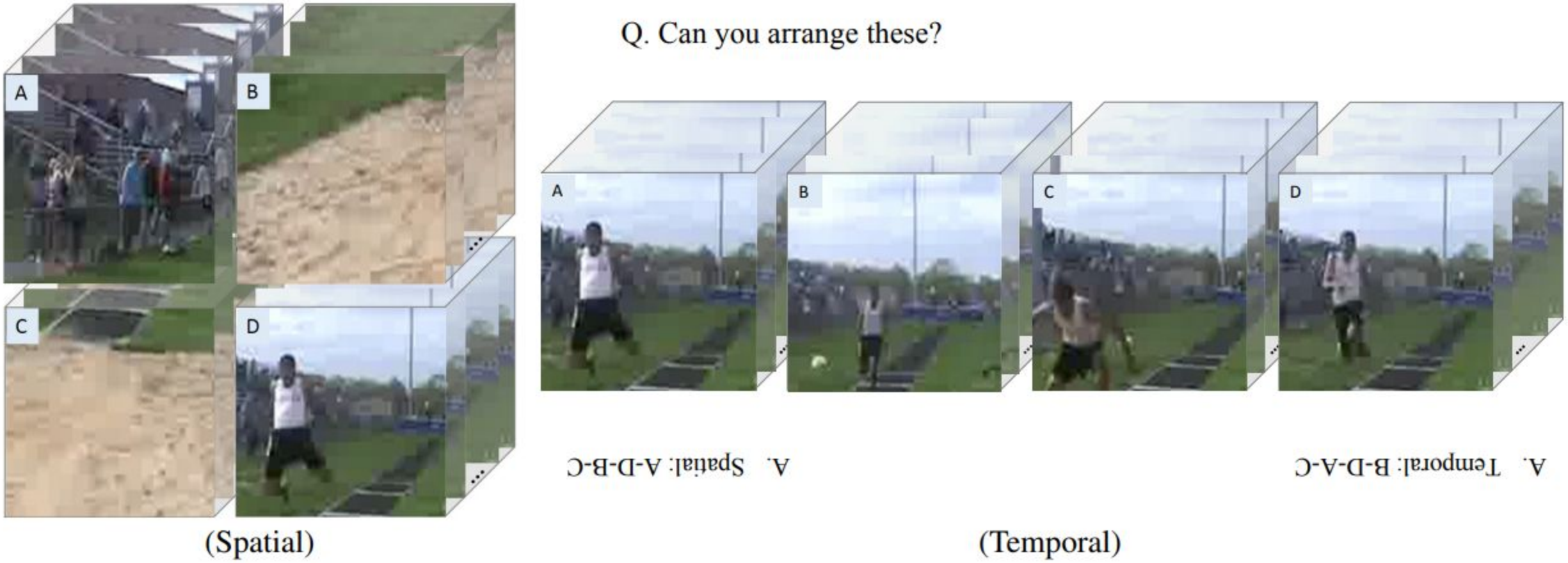
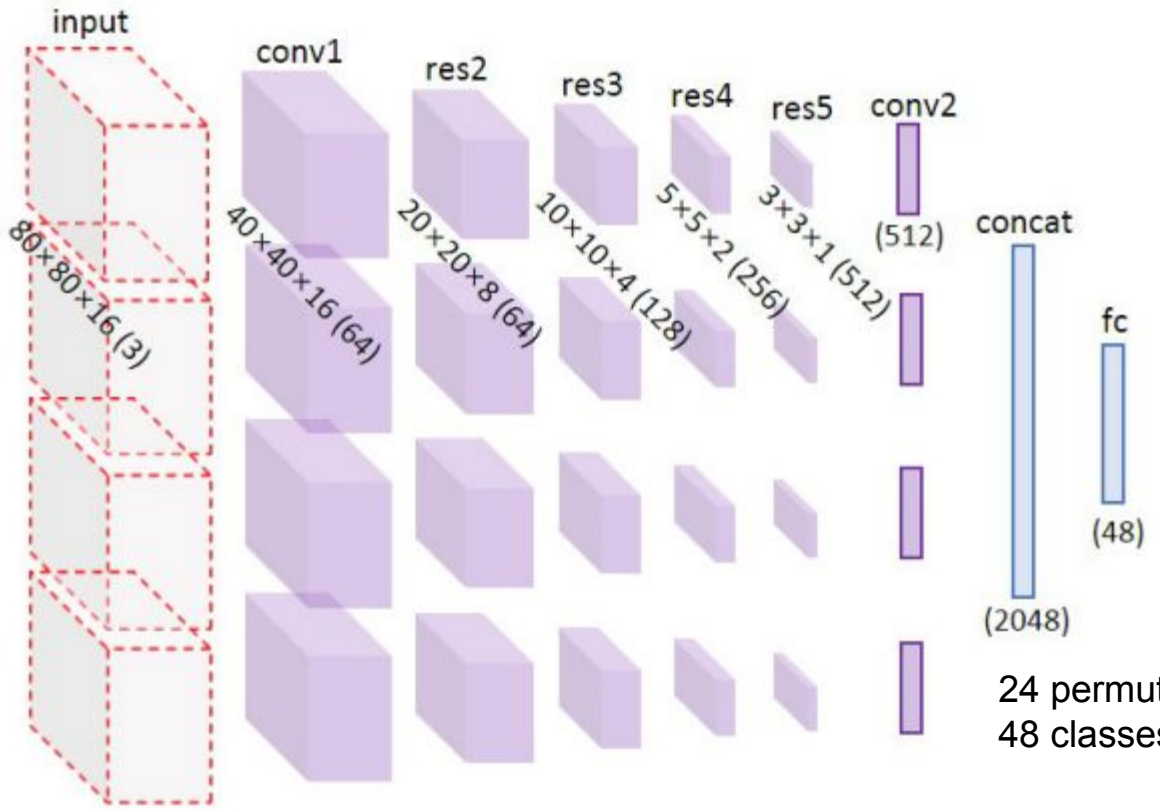


Figure 4: Example spatial and temporal tuples.

Learning Space-Time Features with a 3D CNN

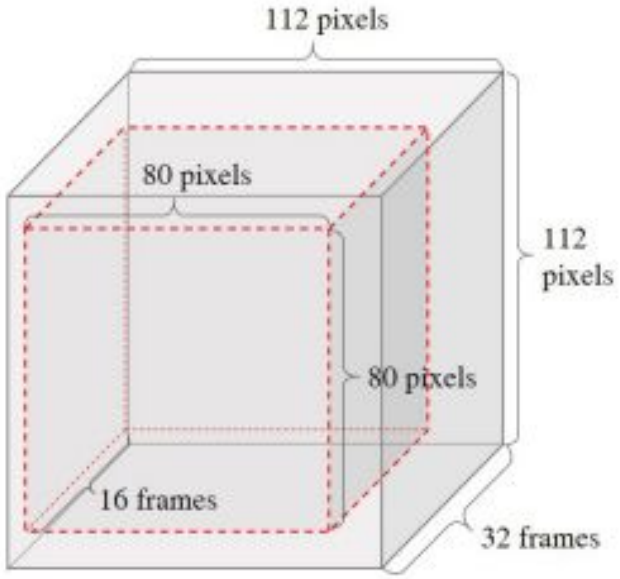


- Four shared 3D CNN towers
- One tower processes one video cube
- Features are combined only at the end
- The network predicts the permutation

24 permutations × 2 orientations =
48 classes

Each tower does not know whether the puzzle is spatial or temporal.

Avoiding Shortcut Solutions



- Possible shortcuts:
- Chromatic aberration
- Matching cube boundaries
- Low-level image statistics

Regularization:

- Channel replication
- Spatio-temporal jittering
- Upside-down classification

Spatial and Temporal Context Are Complementary

Method	UCF101(%)
3D AE	48.7
3D AE + future	50.1
3D inpainting	50.9
3D S-puzzle	58.5
3D T-puzzle	59.3
3D ST score ensemble	61.3
3D ST-puzzle (full)	**65.8**

Table 2: **comparison with alternative methods.** Top-1 accuracies on UCF10. All methods use 3D ResNet-18, and the accuracies are averaged over three splits.

- Joint space-time learning performs better than learning each dimension separately.

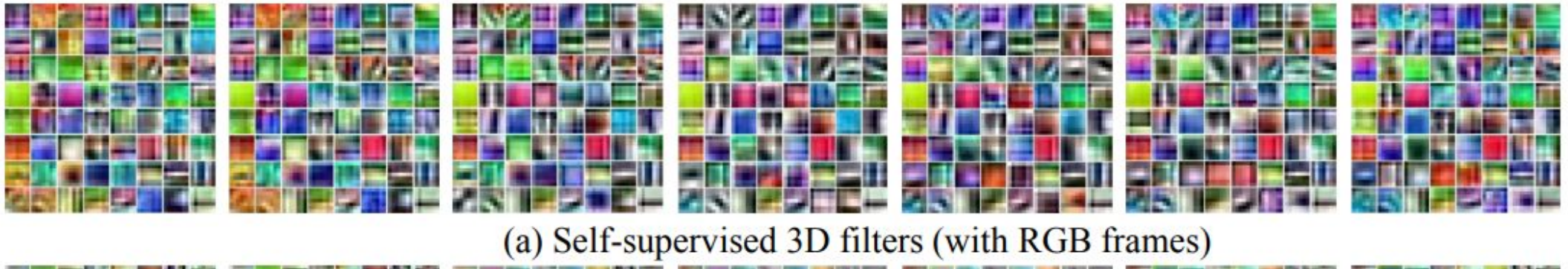
Transfer to Action Recognition

Initialization	UCF101(%)	HMDB51(%)
Random init.	42.4	17.1
3D ST-puzzle (ours)	65.8	33.7
Kinetics 1/8	64.2	33.2
Kinetics 1/4	71.1	41.1
Kinetics 1/2	78.0	48.6
Kinetics full	84.4	56.4
ImageNet-inflated	60.3	30.7

Table 1: **Comparison with random initialization / fully-supervised pretraining.** Top-1 accuracies on UCF101 and HMDB51. All methods use 3D ResNet-18, and the accuracies are averaged over three splits.

- +23.4 points on UCF101
- +16.6 points on HMDB51
- over random initialization

Learned Spatio-Temporal Filters



(a) Self-supervised 3D filters (with RGB frames)



(b) Kinetics-pretrained 3D filters



(c) ImageNet-pretrained 2D filters

Figure 5: **Learned filters with self-supervision vs. fully supervised-pretraining.** Visualization of the learned 64 filters in conv1 layer: (a) the resulting 3D 7×7×7 filters of our self-supervised learning, (b) 3D 7×7×7 filters from the Kinetics-pretrained network, and (c) 2D 7×7 filters from ImageNet-pretrained network. Note that our representation incorporate temporal dynamics and have rich temporal structure, without requiring massive human labels.

Cooperative Learning of Audio and Video Models from Self-Supervised Synchronization

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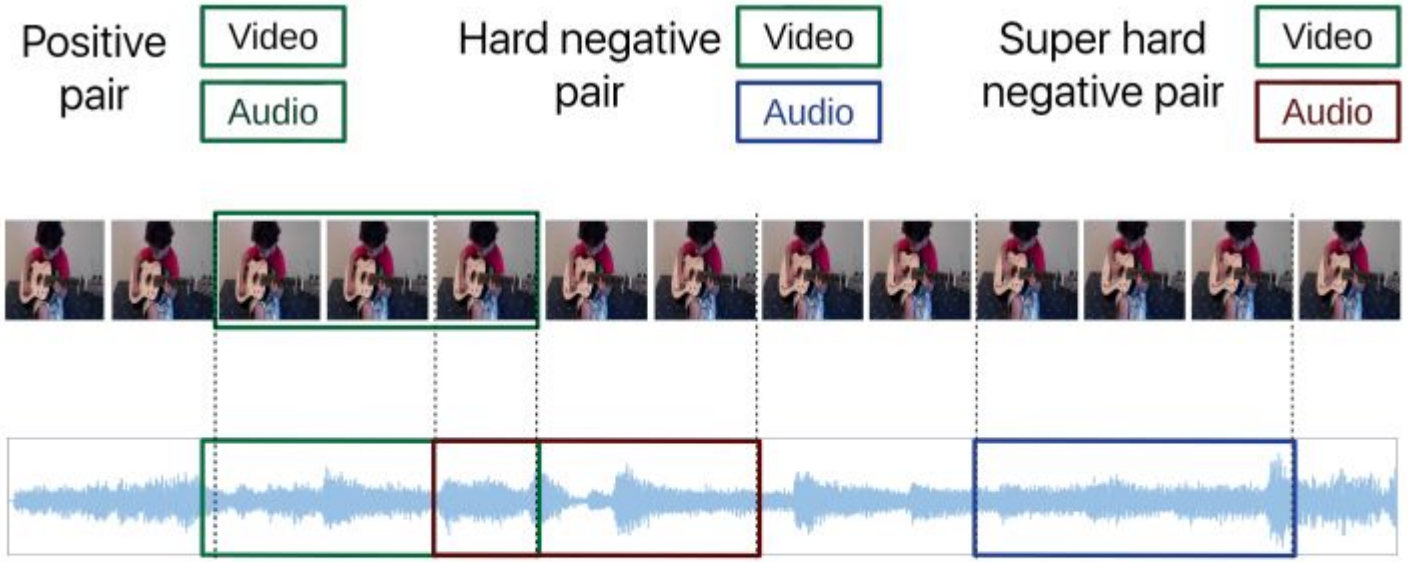
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Korbar et al., NeurIPS 2018, Cooperative Learning of Audio and Video Models from Self-Supervised Synchronization

Audio-Visual Temporal Synchronization

- Video and audio are naturally synchronized.
  - Pretext task:
  - Are the video and audio temporally aligned?



- In-sync pair:
  - Audio and video come from the same time segment.
- Out-of-sync pair:
  - Audio and video come from different time segments.

What must the model understand to detect synchronization?

Cooperative Audio-Video Learning

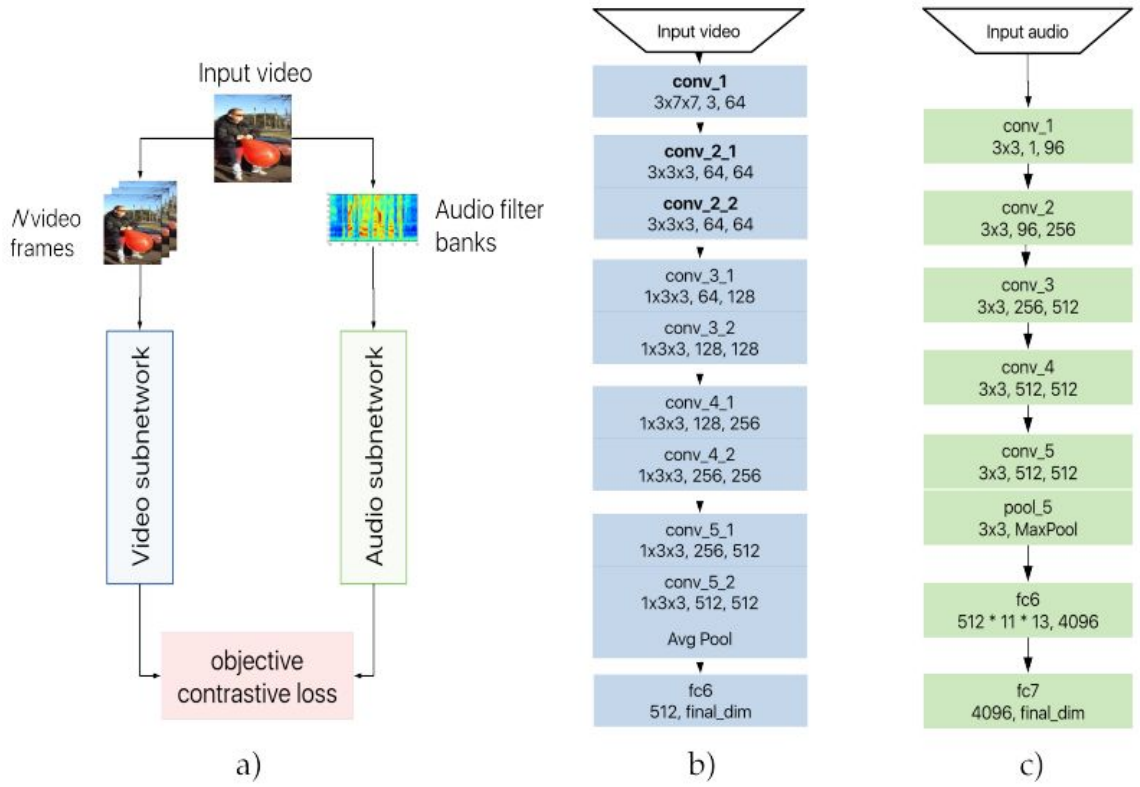


Figure 2: Our architecture design. The complete model for AVTS training can be viewed in (a). The video subnetwork (shown in (b)) is a MCx network [19] using 3D convolutions in the early layers, and 2D convolutions in the subsequent layers. The audio subnetwork (shown in (c)) is the VGG model used by Chung and Zisserman [26].

- Video stream
  - Processes motion and appearance
  - Uses a spatio-temporal CNN
- Audio stream
  - Processes audio filterbank features
  - Learns sound representations

The Choice of Negative Pairs Matters

Curriculum Learning

Table 1: AVTS accuracy achieved by our system on the Kinetics test set, which includes negatives of only "easy" type, as in [21]. The table shows that curriculum learning with a mix of easy and hard negatives in a second stage of training leads to a significant gain in accuracy.

Method	Negative type	Epochs	Accuracy (%)
Single learning stage	easy	1 - 90	69.0
	75% easy, 25% hard	1 - 90	58.9
	hard	1 - 90	52.3
	easy	1 - 50	67.2
Curriculum learning (i.e., second learning stage applied after a first stage of 1-50 epochs with easy negatives only)	75% easy, 25% hard	51 - 90	**78.4**
	hard	51 - 90	65.7

A good pretext task depends not only on the labels,
but also on how positive and negative examples are constructed.

Self-supervised Transformation Learning for Equivariant Representations

Jaemyung Yu[1] **Jaehyun Choi**[1] **Dong-Jae Lee**[1] **HyeongGwon Hong**[1] **Junmo Kim**[1]
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Yu et al., NeurIPS 2024, Self-supervised Transformation Learning for Equivariant Representations

What Do Pretext Tasks Preserve?

- Rotation Prediction
  - The model must preserve rotation information.
- Colorization
  - The model must preserve object and color information.
- Jigsaw Puzzle
  - The model must preserve spatial relationships.

A pretext task determines what information is preserved in the representation.

Should Everything Be Invariant?

- Modern SSL often uses image transformations:
  - Random crop
  - Color jitter
  - Horizontal flip
  - Gaussian blur

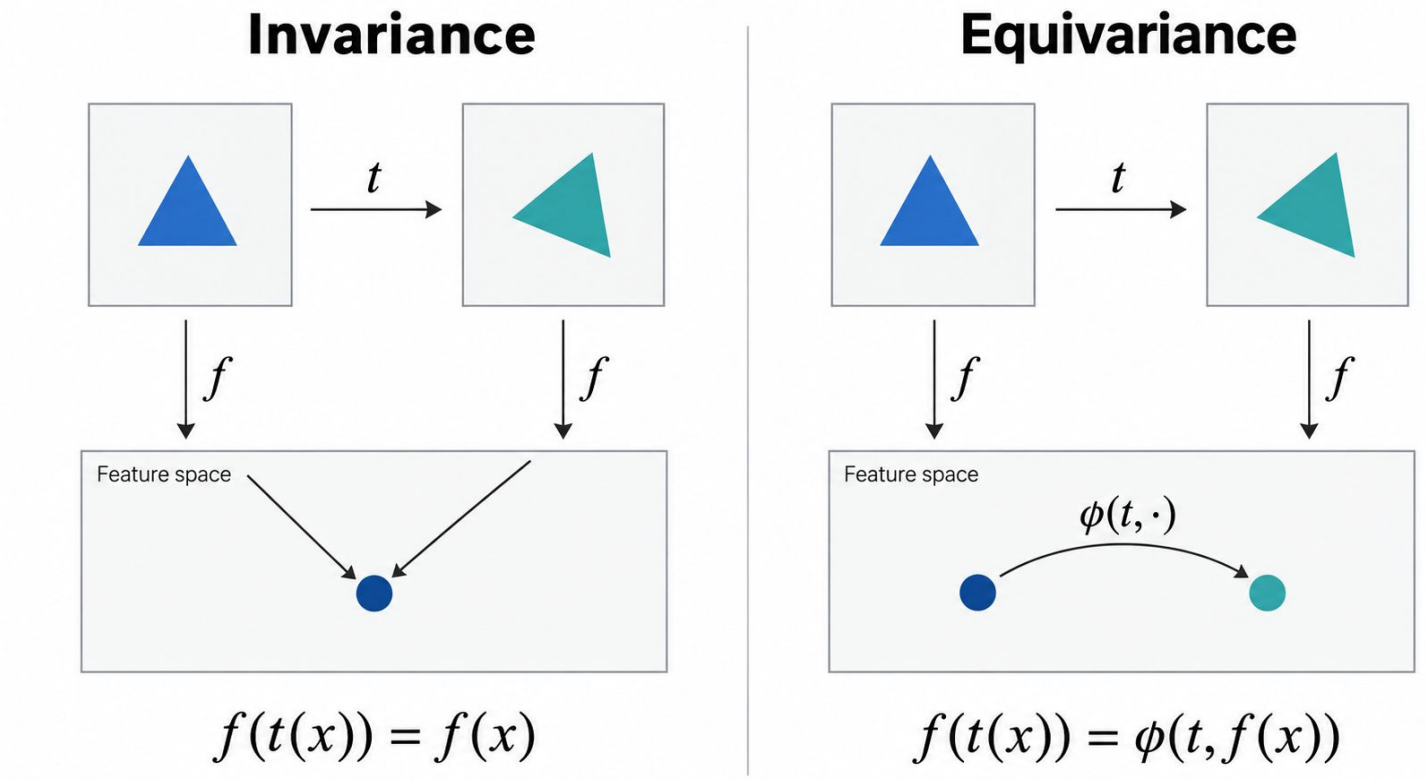
The model is trained to ignore these transformations.

But is invariance always desirable?

Should Everything Be Invariant?

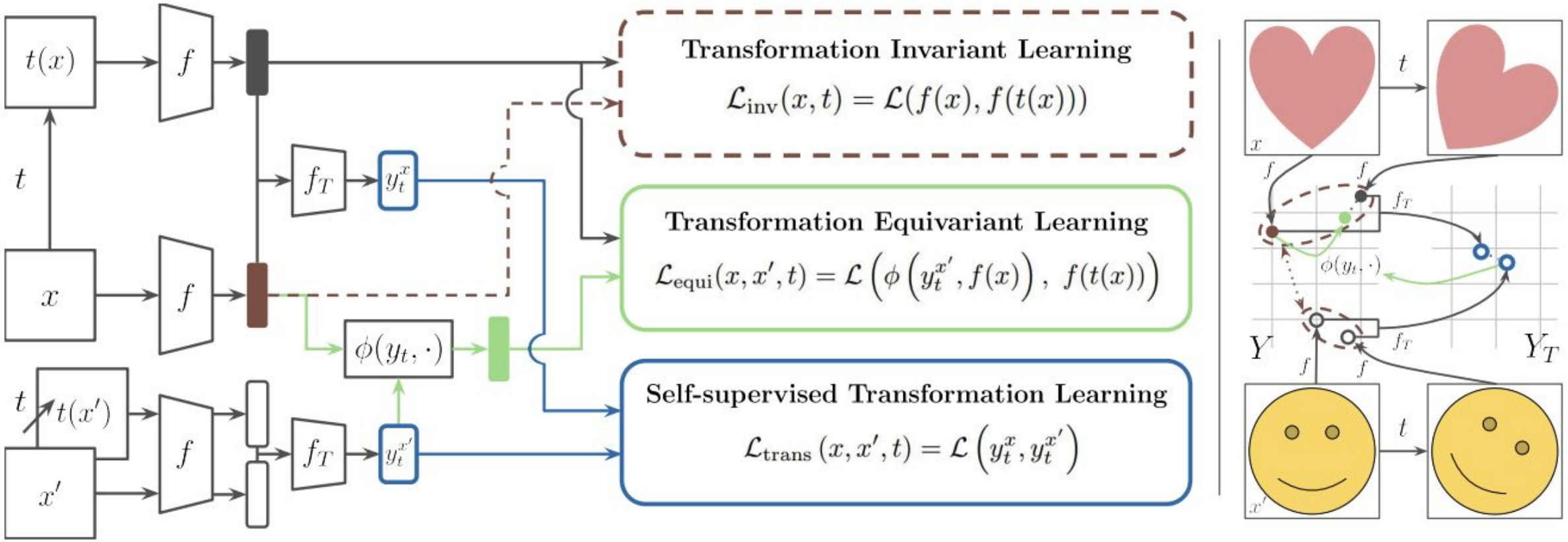
- Classification:
  - “What is the object?” → Invariance may be useful.
- Detection:
  - “Where is the object?” → Spatial information is necessary.
- Flower classification:
  - “What color is the flower?” → Color information is necessary.

Invariance vs. Equivariance



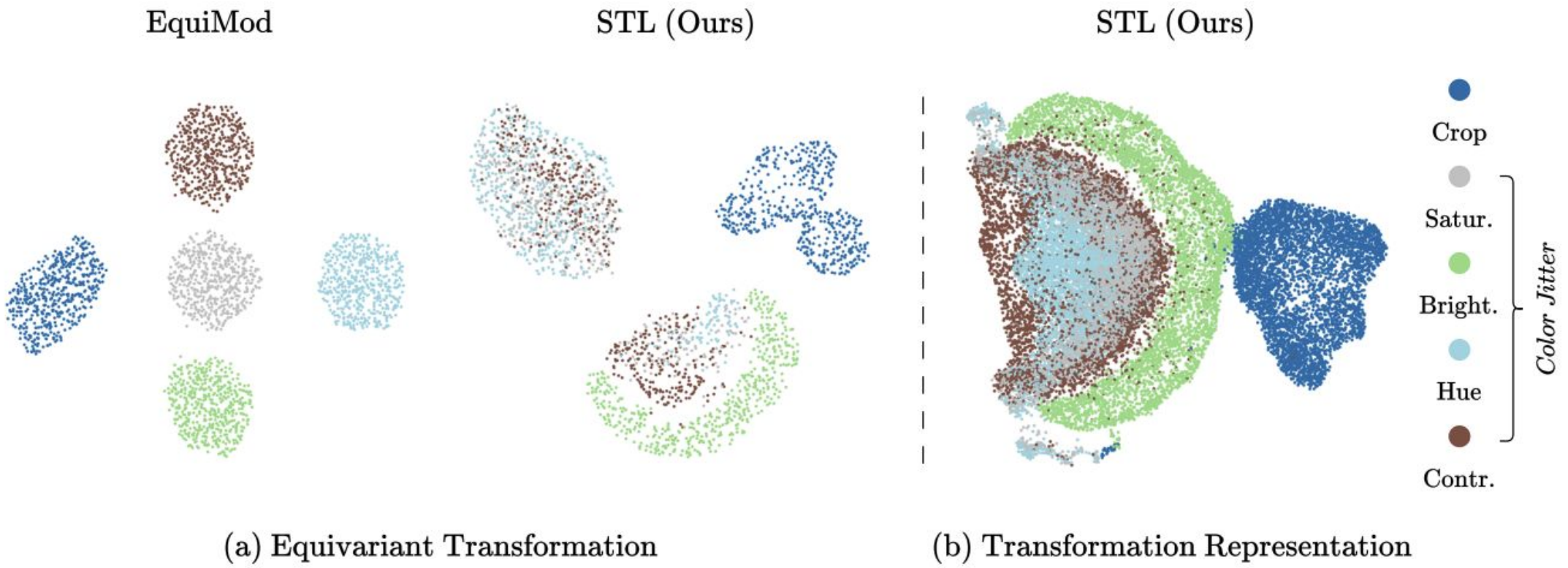
Invariance: representation does not change after the input transformation.
Equivariance: representation changes predictably according to the input transformation.

Transformation Equivariant Learning



Brown: Same image under transformation → Preserve semantic information
Blue: Same transformation on different images → Learn transformation representations
Green: Apply the learned transformation → Predict the transformed representation

What Does STL Learn?

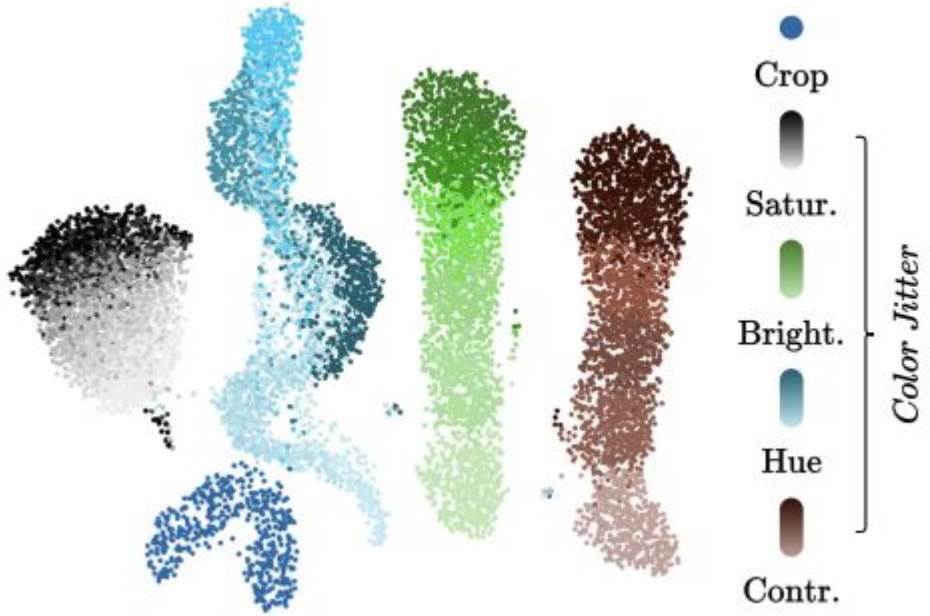


(a) Equivariant Transformation

(b) Transformation Representation

- Crop is separated from color transformations.
- Related color transformations are placed close together.
- This structure is learned without transformation labels.

Transformation Representations by Intensity



UMAP visualization

darker colors representing higher intensities

ResNet-18 model pretrained on STL10
with a transformation backbone

- Different types form different regions.
- Similar intensities are located nearby.
- Intensity changes continuously in the representation space.

STL captures both transformation type and intensity without direct parameter supervision.

Downstream Results

Table 4: **Object Detection.** Evaluation of representation generalizability on a downstream object detection task. Average precision is reported for ImageNet100-pretrained ResNet-50 fine-tuned on VOC07+12.

Method	AP_{all}	AP_{50}	AP_{75}
SimCLR	45.67	72.50	47.83
AugSelf	45.99	72.46	49.23
EquiMod	51.55	78.03	56.17
STL (Ours)	51.95	78.34	56.96
with AugMix (Ours)	**52.70**	**78.81**	**57.76**

Table 5: **Transformation Prediction.** Evaluation of transformation representation from learned represetation pairs. Regression tasks use MSE loss, and transformation type classification uses accuracy (%).

	Regression (↓)			Classification (↑)
Method	Crop	Color	All	Trans. Type
SimCLR	0.02	0.13	0.08	68.54
AugSelf	**0.01**	0.04	0.03	88.49
EquiMod	**0.01**	0.07	0.04	82.20
STL (Ours)	**0.01**	**0.03**	**0.02**	**93.67**

Preserving transformation information improves localization and generalization.

Next Week..

- Contrastive Learning